ISA Data Fluency

IMBA Session 3

"There is magic in graphs. The profile of a curve reveals in a flash a whole situation – the life history of an epidemic, a panic, or an era of prosperity. The curve informs the mind, awakens the imagination, convinces." –Henry D. Hubbard

Professor Christina Stathopoulos, September 2026

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DATAcated
Book of the Day
ColorWise by Kate Strachnyi
O'REILLY
ColorWise
A DATA STORYTELLER'S GUIDE TO THE INTENTIONAL USE OF COLOR
KATE STRACHNYI
ColorWise
ie
University
Higher Education

Session 3 Agenda

1

Color Theory

2

Data → to → Viz

(time-permitting, we will cover part of this and the remainder will be assigned as reading homework)



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Review

Marks?	Basic visual objects that represent data
Channels?	Visual variables that we can use to represent characteristics of these marks
Pre-Attentive Attributes?	Pattern recognition, attributes that our brain unconsciously focuses on
Gestalt Principles?	Making sense of things by seeing a whole, rather than the individual parts

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Color Theory

The Intentional Use of Color



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George Firican 💡 • 1st
Award Winning Data Governance Leader | Founder of LightsOnData | Po...
3d •

Zoom in to see what camouflage really means.
And remember, in data visualizations...

...don't camouflage the visuals and the message you want to get across.

Use some of these things to draw your user's attention:
💡 Preattentive attributes
💡 Annotations
💡 The Z layout pattern

and remember that less is more.

If you'd like to learn more about Data Visualization Best Practices for Data Storytelling checkout our course with the same name available on LightsOnData dot com.

Did you see the planes in the photo?

#data #dataviz #datastorytelling #datavisualization #datascience





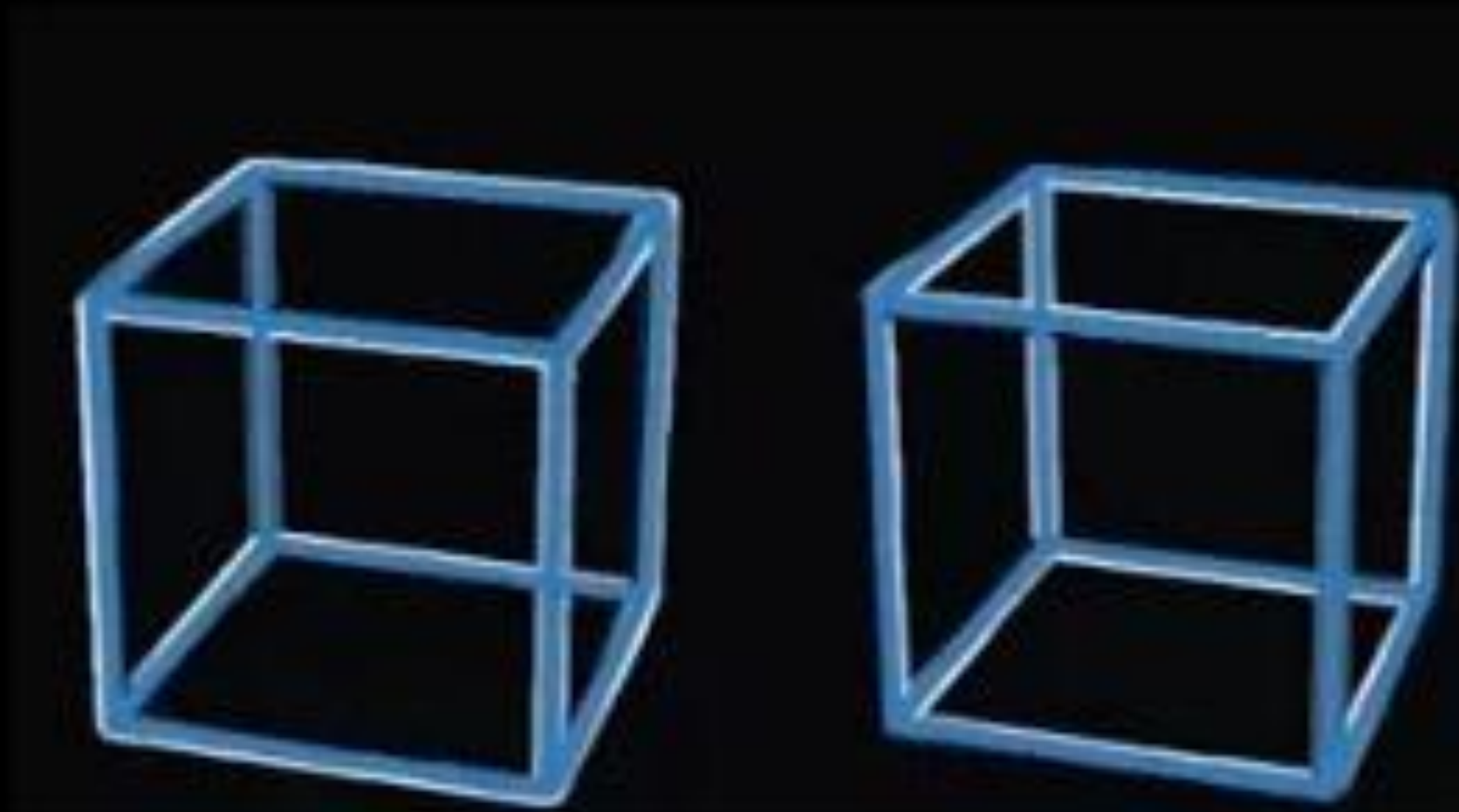
"..remember, in data visualizations, don't camouflage the visual and the message you want to get across."

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Static or Rotating?

Optical Illusion of 'Moving' Cubes Has Stumped The Internet



When Visualizing Data…



it’s not just about choosing the right graphs.

Choose the right colors too!

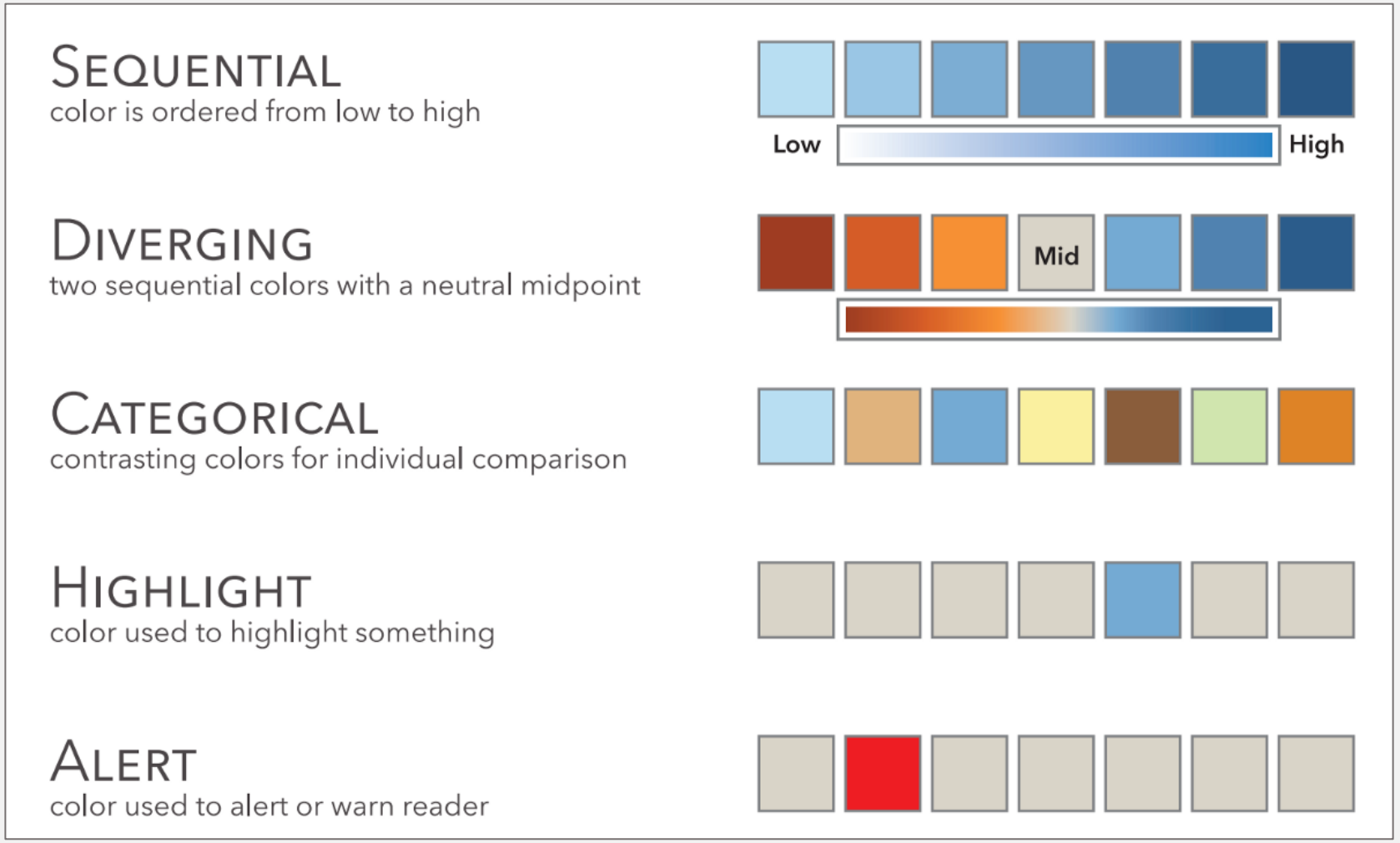
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HOW TO USE
COLORS

Source: Steve Wexler, Jeffrey Shaffer & Andy Cotgreave. The Big Book of Dashboards (2017)

Types of Colors Used in Data Viz



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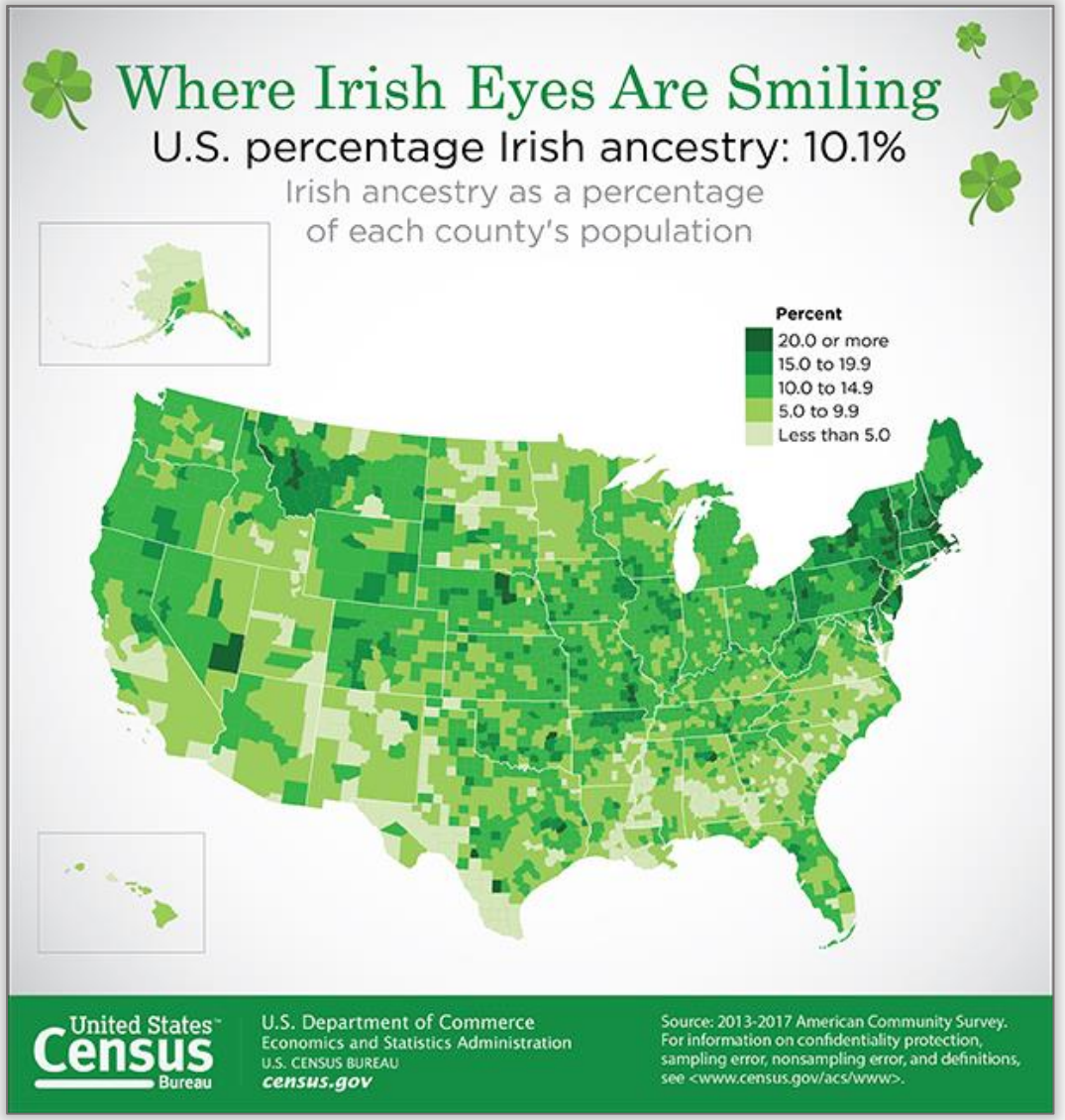
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Types of Colors Used in Data Viz

SEQUENTIAL
color is ordered from low to high



Use **Sequential** for the most intuitive reading of a numeric, color scale.

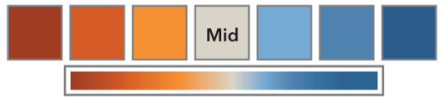


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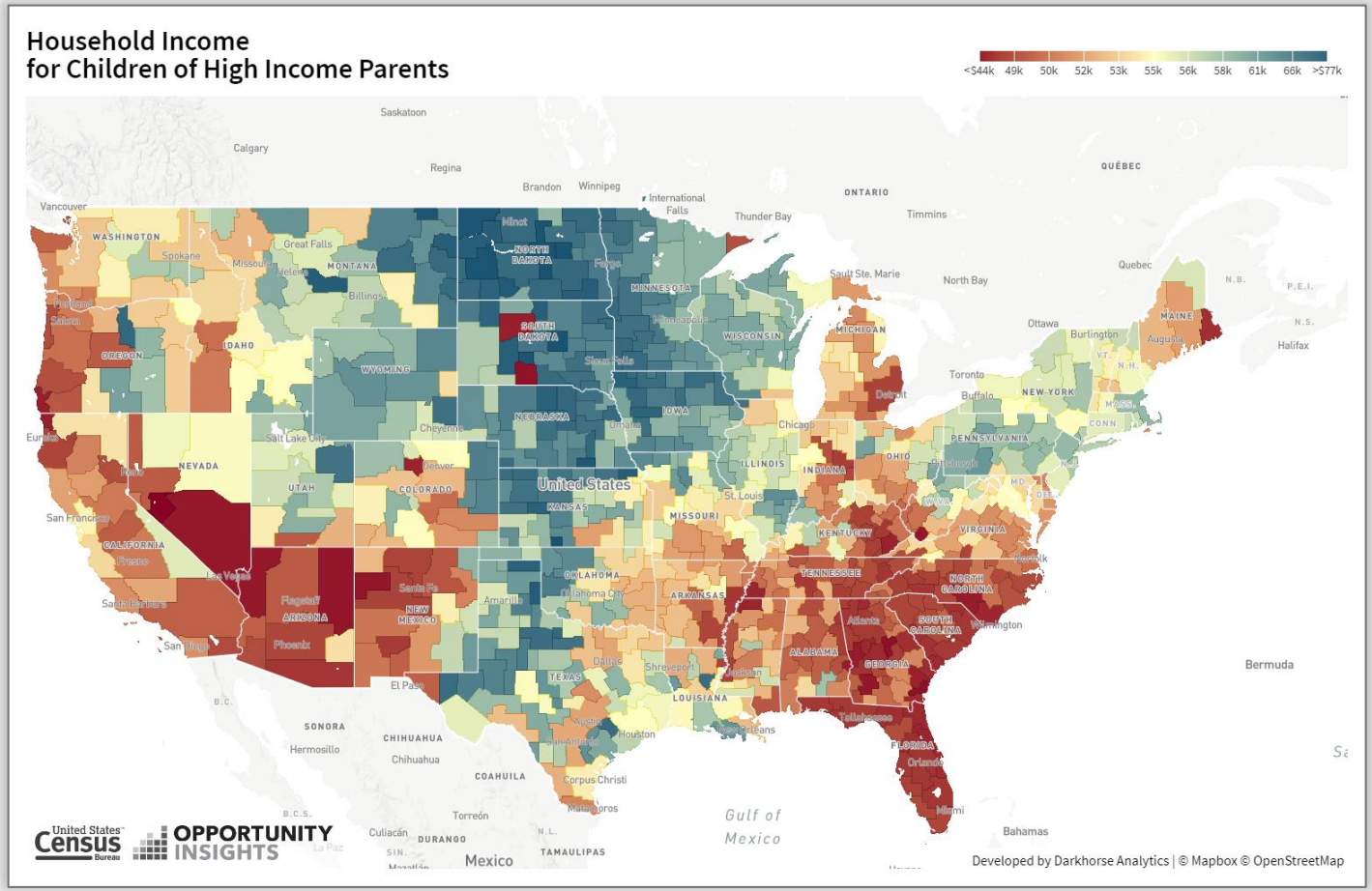
Types of Colors Used in Data Viz

DIVERGING
two sequential colors with a neutral midpoint



Use **Diverging**…

- If there is a meaningful middle point.
- To highlight varying extremes.
- To emphasize the difference between data points.



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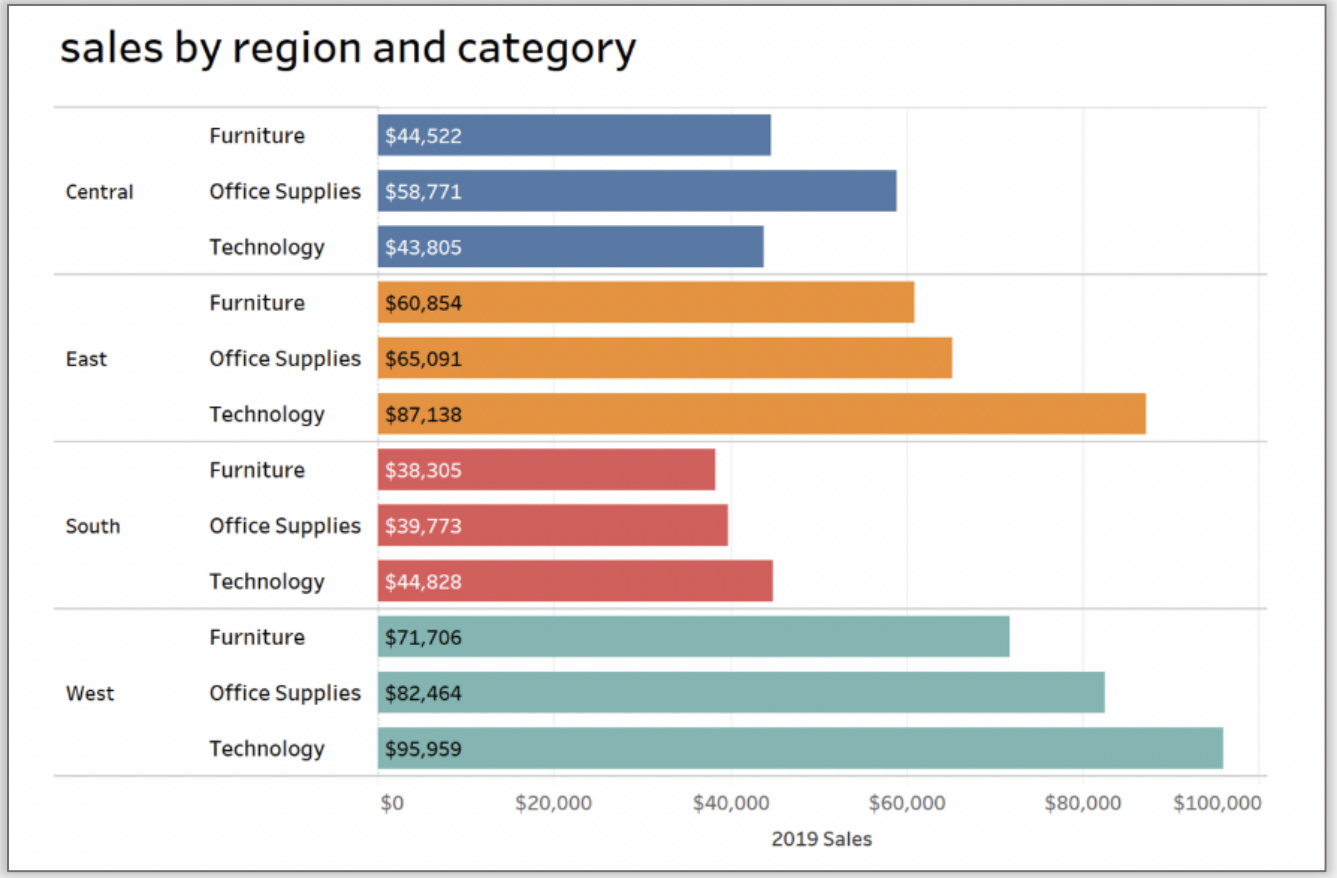
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Types of Colors Used in Data Viz

CATEGORICAL
contrasting colors for individual comparison



Use **Categorical** for clear differentiation of categorical variables.



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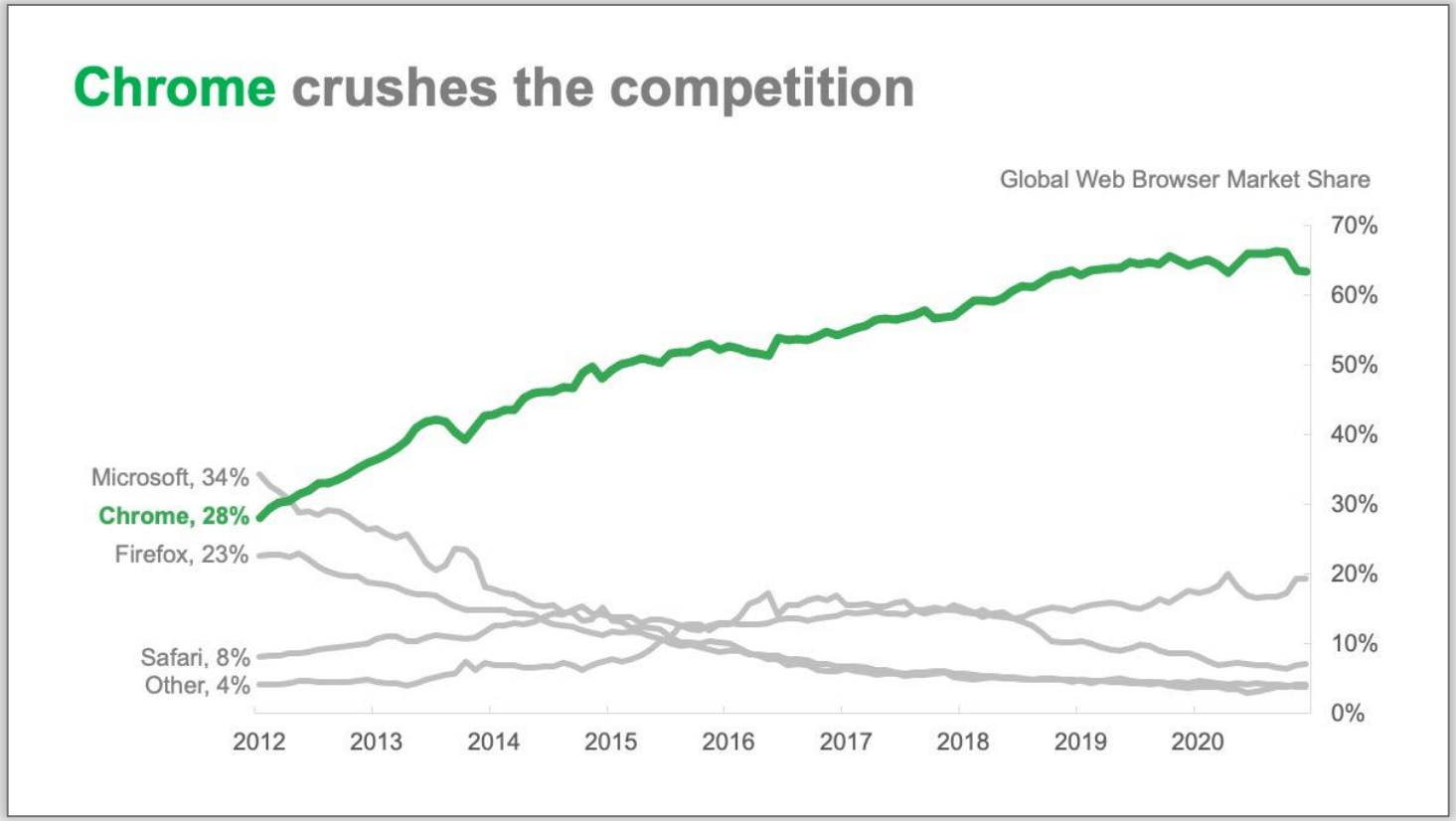
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Types of Colors Used in Data Viz

HIGHLIGHT
color used to highlight something



Use **Highlight** to make your viewers focus on a specific piece.



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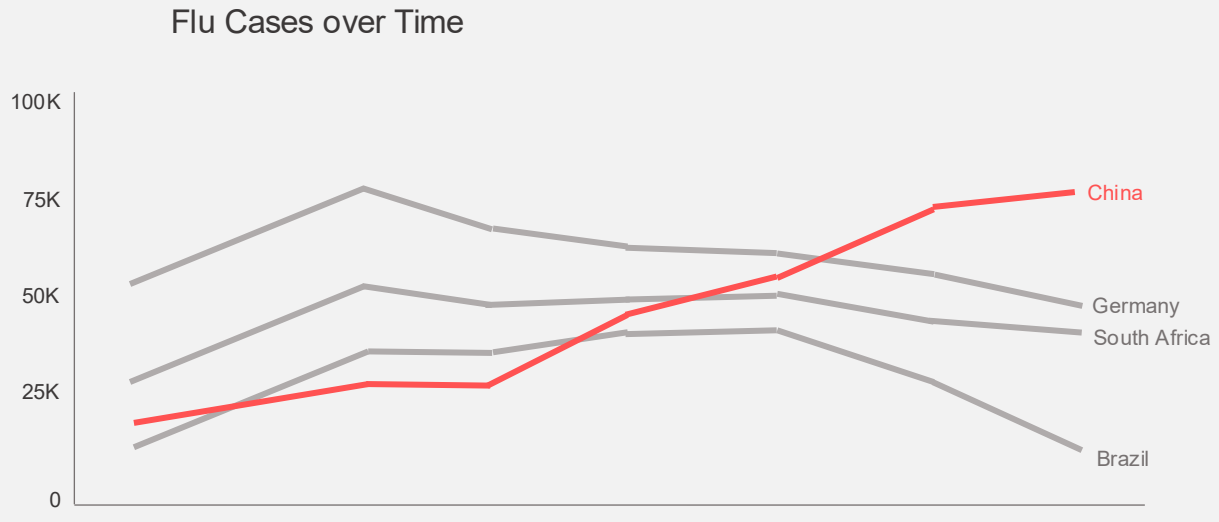
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Types of Colors Used in Data Viz

ALERT
color used to alert or warn reader



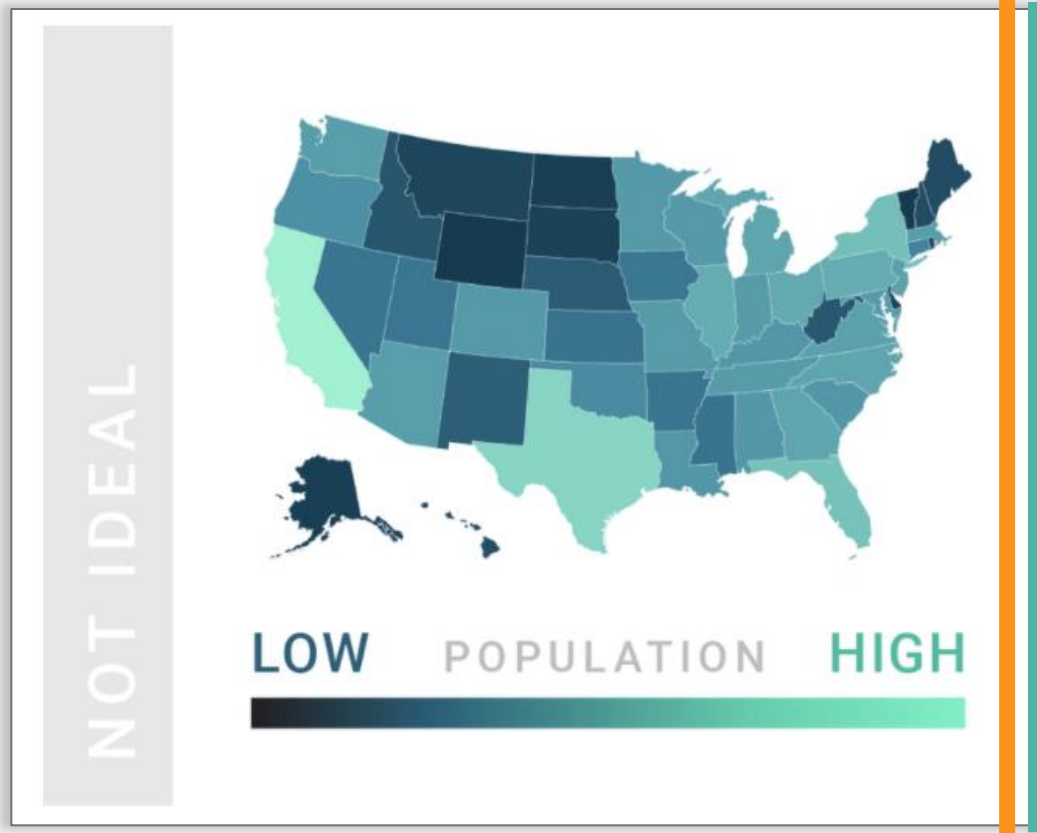
Use **Alert** to warn your viewers of something, typically a negative connotation displayed with the color red.

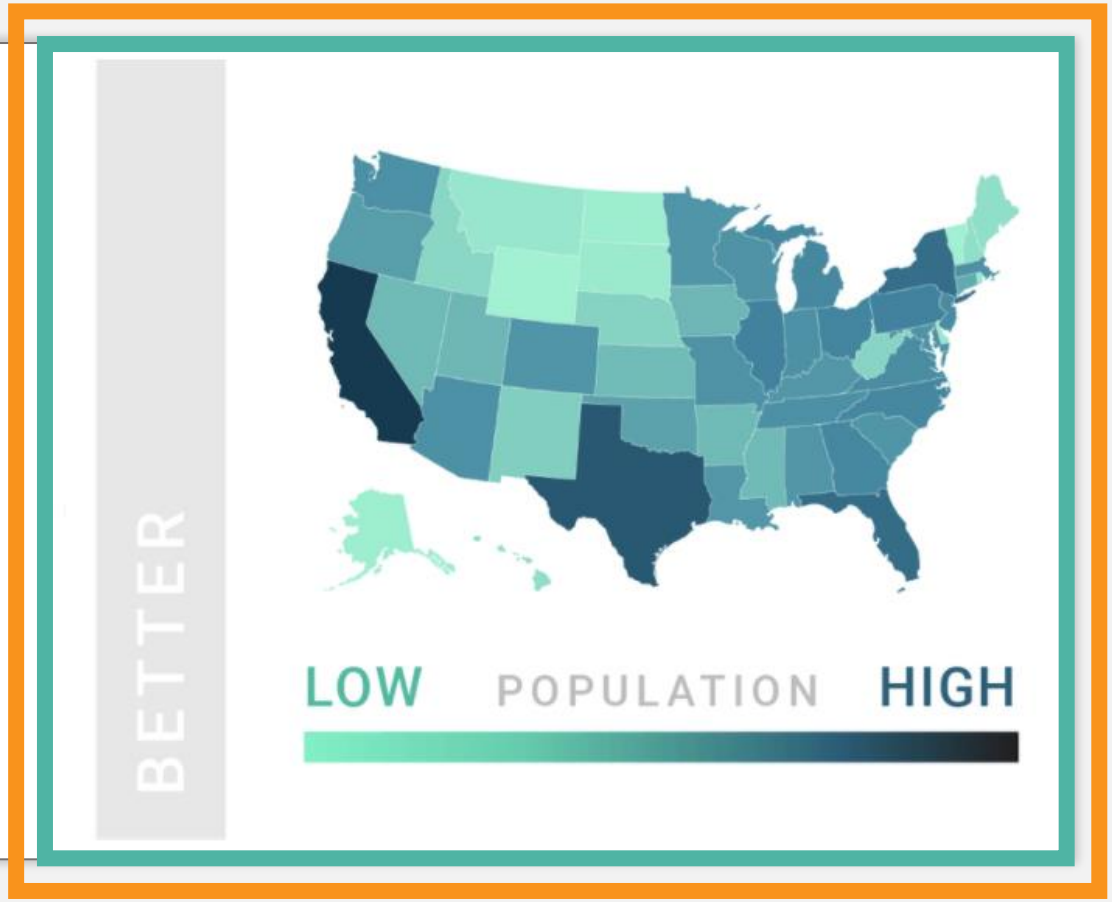


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DO – Use Light to Dark for Continuous Data



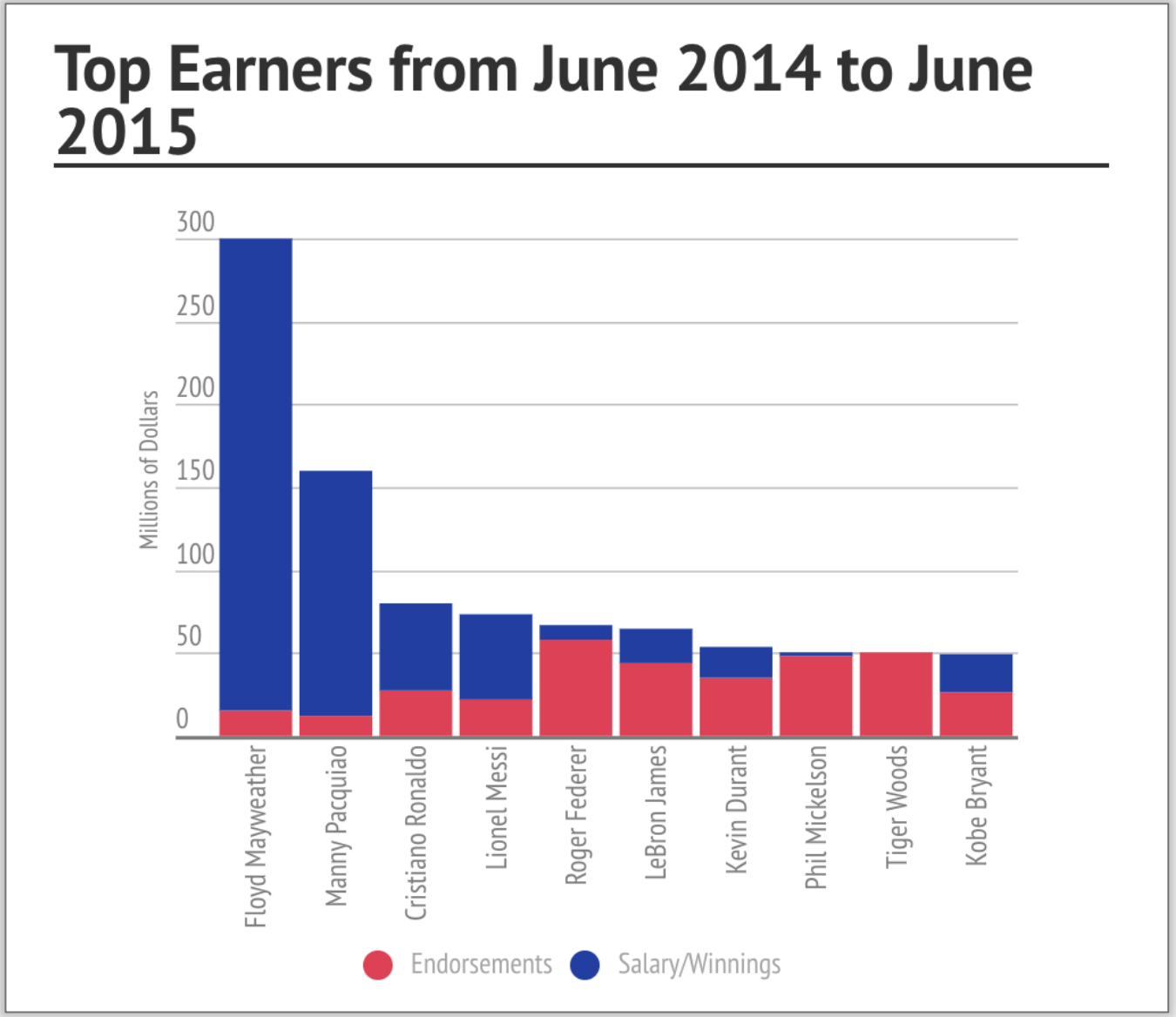


DATAQUEST

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DO – Use Contrasting Colors for Comparison

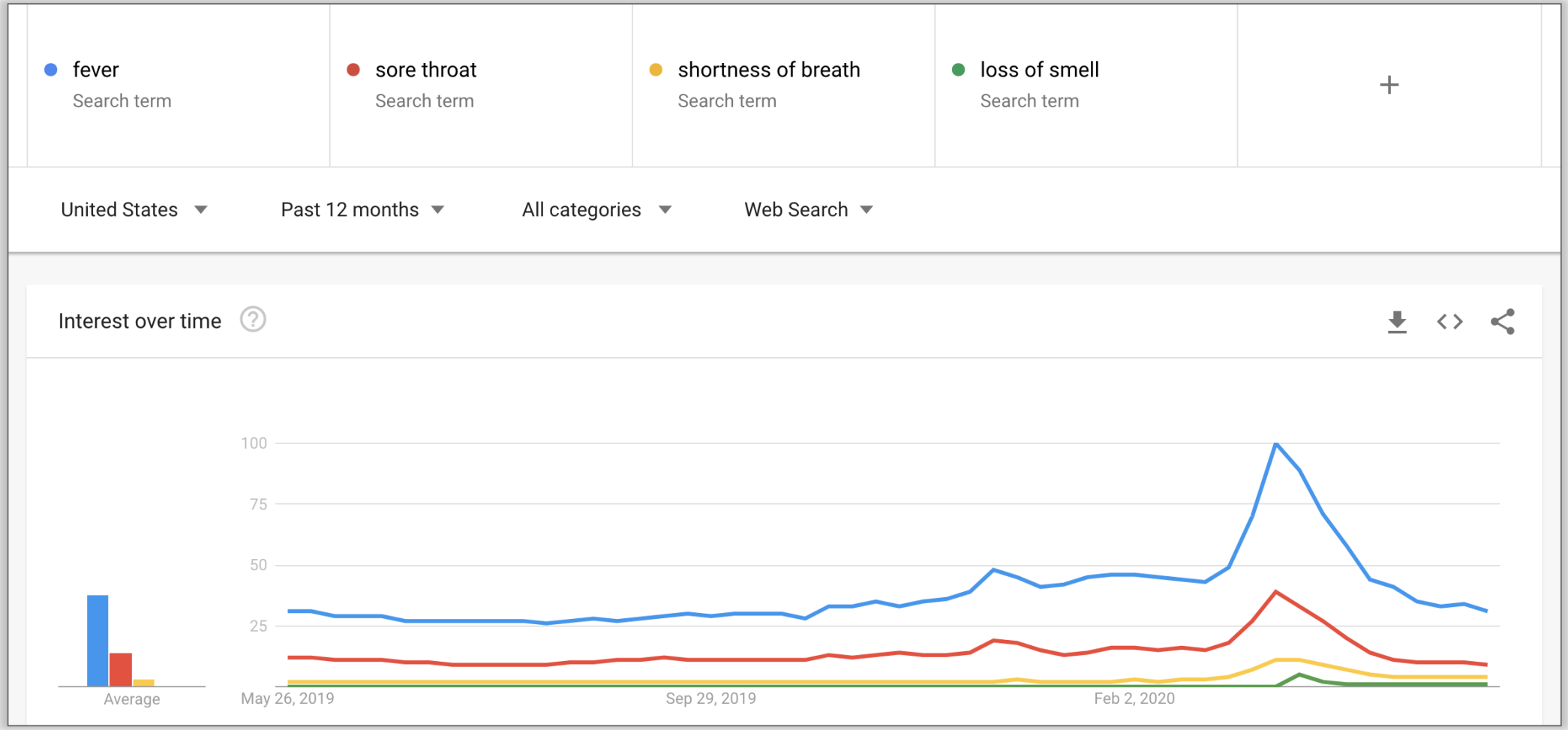


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DO – Use your Brand Colors



Google Trends

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DO – Include a Color Key (Legend)



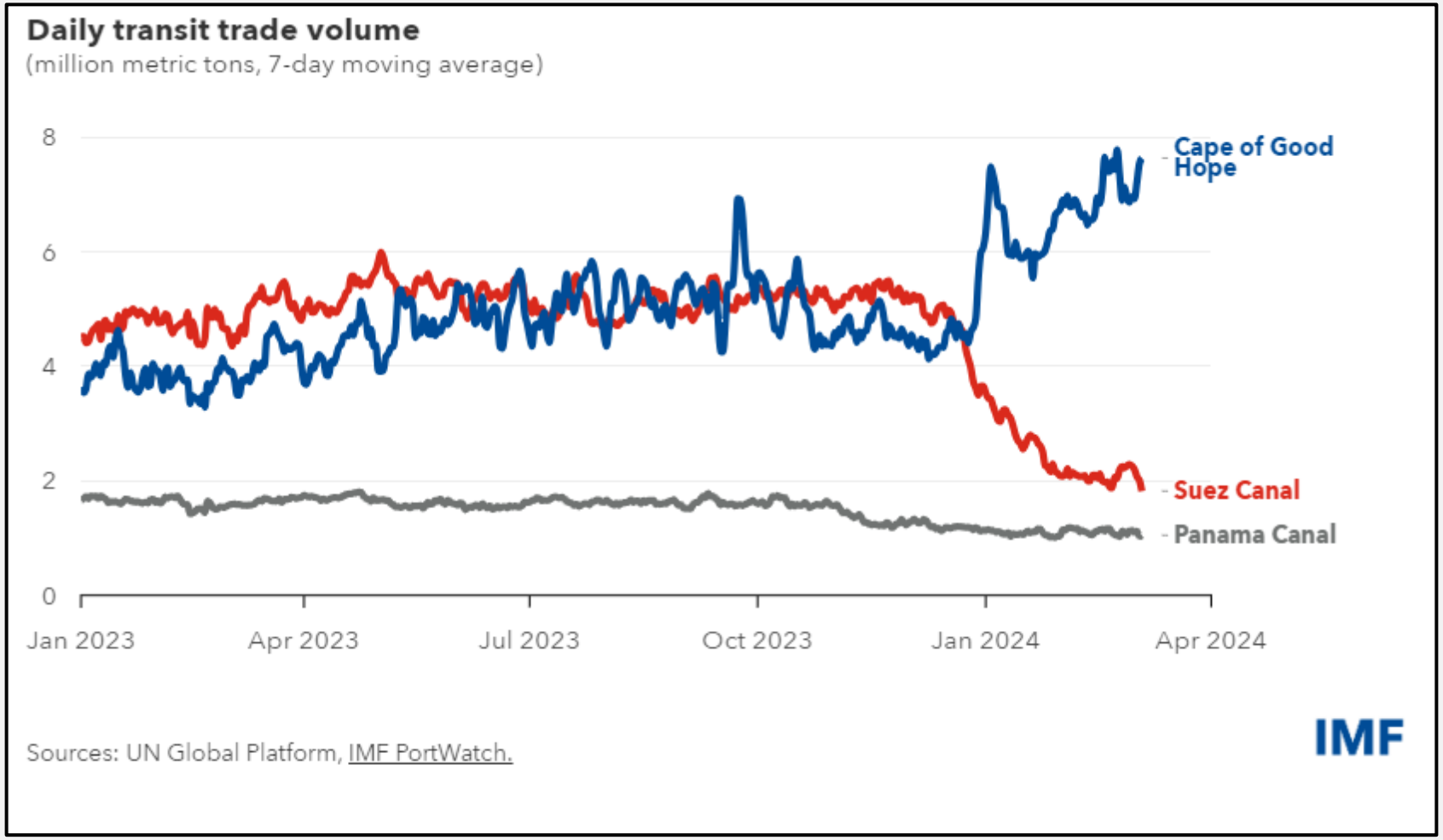
DATAQUEST

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DO – Include a Color Key (Legend)

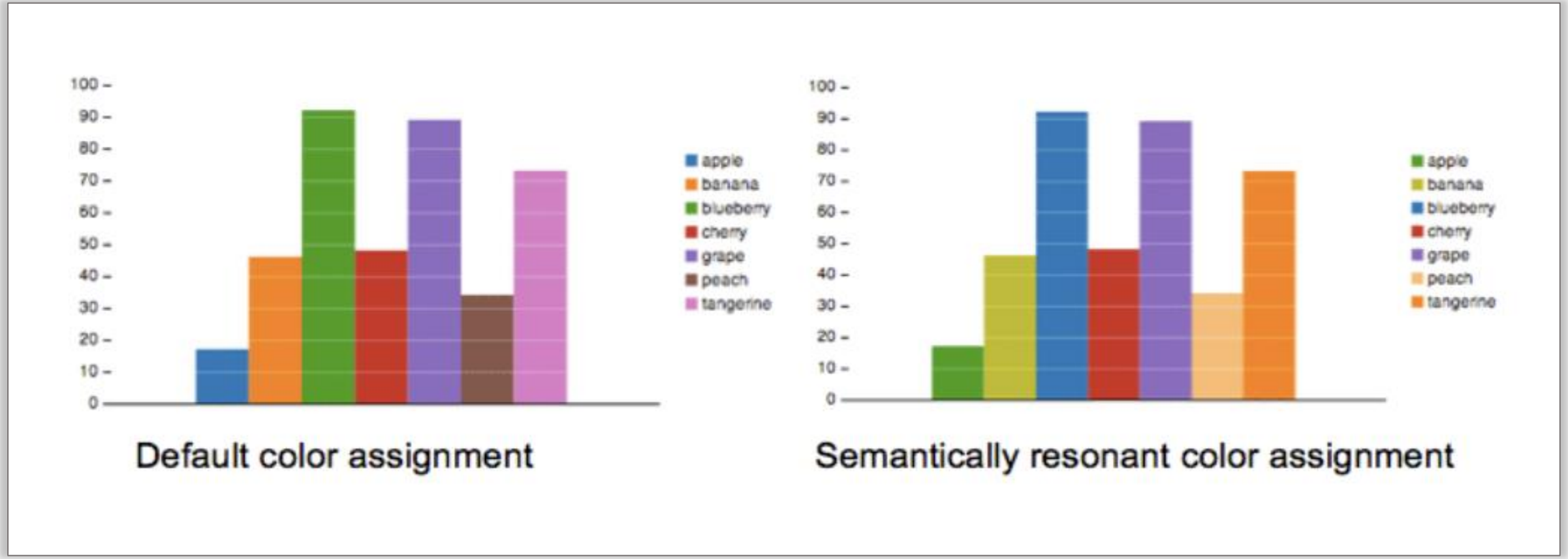
(can be added directly on the graph too)



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DO – Use Semantic Color Association (when applicable)

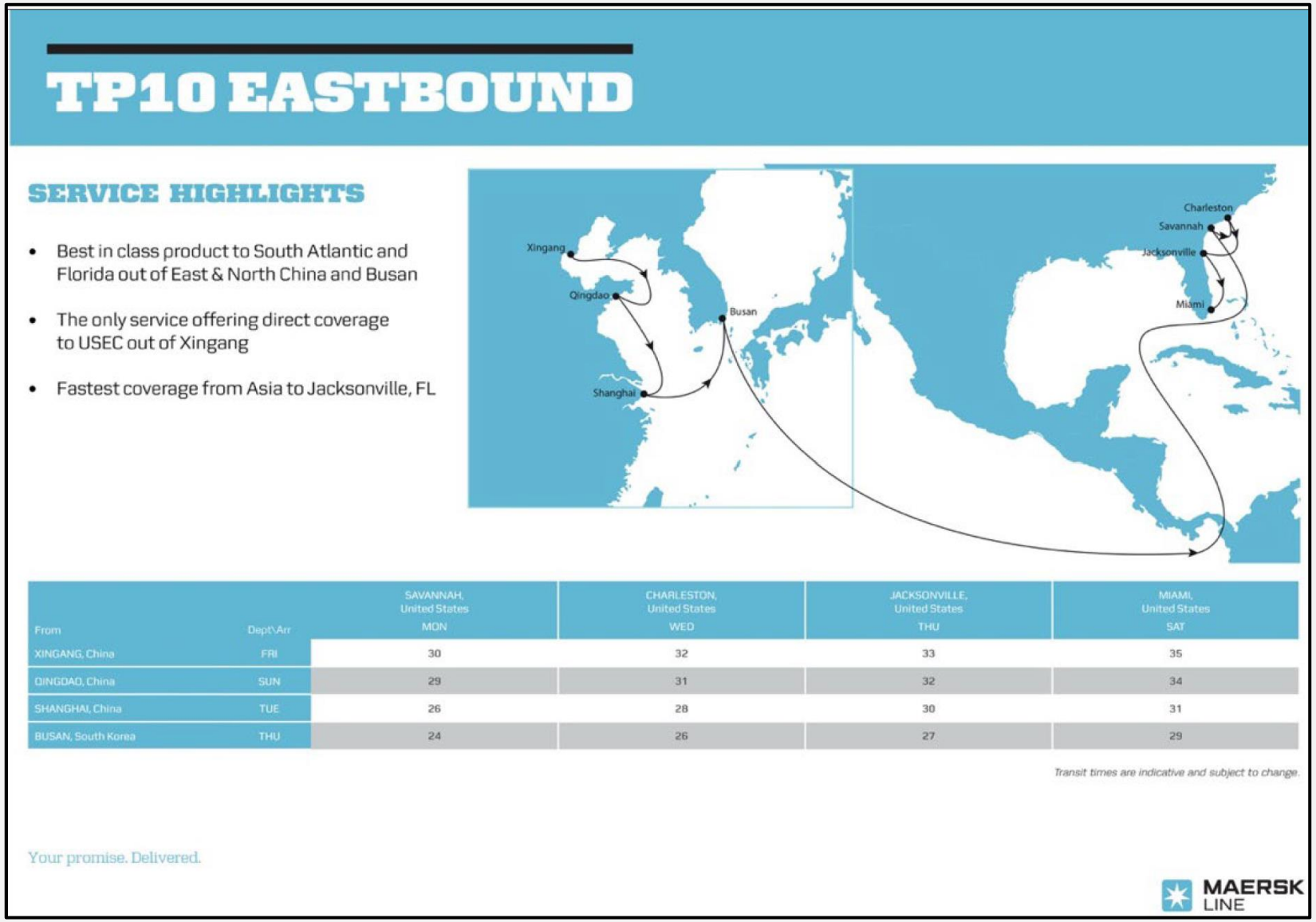


infogram

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DO – Use Semantic Color Association (when applicable)



TP10 EASTBOUND

SERVICE HIGHLIGHTS

- Best in class product to South Atlantic and Florida out of East & North China and Busan
- The only service offering direct coverage to USEC out of Xingang
- Fastest coverage from Asia to Jacksonville, FL

From	Dept\Arr	SAVANNAH, United States MON	CHARLESTON, United States WED	JACKSONVILLE, United States THU	MIAMI, United States SAT
XINGANG, China	FRI	30	32	33	35
QINGDAO, China	SUN	29	31	32	34
SHANGHAI, China	TUE	26	28	30	31
BUSAN, South Korea	THU	24	26	27	29

Transit times are indicative and subject to change.

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MAERSK LINE

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DO – Consider Color Psychology



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Source: Six Degrees. https://www.six-degrees.com/pdf/International-Color-Symbolism-Chart.pdf

DO – Consider Color Psychology



INTERNATIONAL COLOR SYMBOLISM

BLACK

WESTERN
death, evil, sin, funerals, mourning, rebellion, elegance, sophistication, formality, power, control

EASTERN
wealth, prosperity, health, masculinity, knowledge, mystery, evil, mourning

EUROPE
mourning, death, evil, formality, elegance, sophistication

MIDDLE EAST
rebirth, evil, mystery

AFRICA
age, wisdom, maturity, masculinity

BRAZIL
sophistication, authority, mourning, religion, formality

CHEROKEES
problems, death, represents the West

CHINA
trust, high quality, color for young boys

INDIA
laziness, anger, intolerance, apathy, evil, darkness, negativity, lack of desirability

ITALY
death, mourning, humility

JAPAN
night, unknown, nonbeing, mystery, anger

S. KOREA
darkness, mystery, mourning

THAILAND
unhappiness, bad luck, evil

AUSTRALIAN ABORIGINES
ceremonial color

WHITE

WESTERN
purity, peace, elegance, wedding color, brides, angels, doctors, antiseptic

EASTERN
death, mourning, funerals, sadness, purity, age, unhappiness, misfortune

EUROPE
purity, cleanliness, emptiness, neutrality, antiseptic, surrender

MIDDLE EAST
purity, mourning

AFRICA
victory, purity

CHEROKEES
peace, happiness, represents the South

CHINA
death, mourning, virginity, purity, humility, age, misfortune

INDIA
creation, rebirth, light, serenity, reincarnation, peace, purity, unhappiness, widow, death, funerals

IRELAND
leisure, sports, peace

ITALY
virtue, purity

JAPAN
death, mourning

S. KOREA
purity, innocence, morality, birth & death, patriotism

THAILAND
purity, auspiciousness

YELLOW

WESTERN
happiness, joy, hope, optimism, creativity, energy, awareness, hazards, warning, weakness, femininity

EASTERN
sacredness, sovereignty, earth, power, royalty, sun, happiness, masculinity

EUROPE
happiness, joy, quality, weakness, hazard, warning

MIDDLE EAST
happiness, prosperity

AFRICA
wealth, high rank, religion, ceremony, visibility

AUSTRALIA
resurrection, rebirth

CHINA
sacredness, sovereignty, nourishment, royalty, honor, masculinity

EGYPT
soul, sun, happiness, prosperity, mourning

FRANCE
summer, joy, jealousy

GREECE
sadness

INDIA
merchants, sacredness, auspiciousness, peace, happiness, meditation, mental development

JAPAN
courage, nobility, beauty, refinement, aristocracy, cheerfulness

MEXICO
mourning

NETHERLANDS
food

SAUDI ARABIA
strength, reliability

THAILAND
Buddhism, royalty, auspiciousness, color for Monday

GREEN

WESTERN
nature, spring, new birth, regeneration, luck, go, environmental awareness, money, jealousy, greed, Saint Patrick's Day

EASTERN
eternity, new life, regeneration, family, health, prosperity, peace

EUROPE
nature, fertility, confidence, jealousy, inexperience

MIDDLE EAST
islam, strength, fertility, luck, wealth, prestige

CHINA
spring, youth, birth, infidelity, exorcism, desirability, disgrace

EGYPT
hope, spring, fertility

INDIA
islam, hope, new beginnings, nature, harvest, virtue, happiness

INDONESIA
a forbidden color

IRELAND
Ireland – the Emerald Isle, patriotism, nationalism

ISLAM
perfect faith

ISRAEL
bad news

JAPAN
eternal life, youth

S. KOREA
youth, energy

NORTH AFRICA
corruption

SAUDI ARABIA
wealth, prestige

SOUTH AFRICA
death

THAILAND
color for Wednesday

BLUE

WESTERN
trust, loyalty, authority, conservatism, business, peace, calm, depression, sadness, masculinity

EASTERN
immortality, wealth, self-cultivation

EUROPE
truth, responsibility, fidelity, serenity

MIDDLE EAST
safety, protection

BELGIUM
traditionally used for baby girls

CHEROKEES
defeat, trouble

CHINA
immortality, femininity

EGYPT
truth, justice, virtue, faith, protection, reproduction

FRANCE
water, reliability, trust

INDIA
heavens, love, truth, mercy, strength, bravery, manliness, determination

INDONESIA
sadness

IRAN
heaven, spirituality, mourning, immortality

ISRAEL
coat of arms

ITALY
mourning, heaven, purity

S. KOREA
integrity, mourning

MEXICO
trust, serenity, mourning

THAILAND
color for Friday

TURKEY
healing, wealth, repels evil

PURPLE

WESTERN
royalty, spirituality, faith, wealth, fame, high-ranking positions of authority, luxury, calm, magic

EASTERN
wealth, nobility, privilege, sorrow, mourning

EUROPE
royalty, nobility, luxury, power, vanity

MIDDLE EAST
wealth

BRAZIL
death, mourning

EGYPT
virtue, faith

FRANCE
freedom, peace

INDIA
sorrow, unhappiness

ITALY
death, mourning, endurance of suffering, nobility

JAPAN
wealth, privilege, power, royalty

THAILAND
death, mourning, widows, color for Saturday

PINK

WESTERN
love, romance, femininity, childhood, babies, nurturing, sweetness, Valentine's Day

EASTERN
marriage, femininity

EUROPE
baby girls, delicate, flirtation, sensitivity, serenity, femininity

BELGIUM
traditionally used for baby boys

JAPAN
spring, femininity, youth, good health, well-liked by both males & females

S. KOREA
trust

THAILAND
color for Tuesday

RED

WESTERN
stop, danger, anger, blood, energy, excitement, action, adventure, love, passion, Valentine's Day

EASTERN
prosperity, good fortune, happiness, worn by brides, celebration, communism

EUROPE
visibility, cheapness, loudness

MIDDLE EAST
danger, caution, evil

AFRICA
death, mourning

CHINA
good fortune, happiness, celebration, summoning, vitality, prosperity, long life, wedding color

ENGLAND
power, authority, government

FRANCE
passionate love, lust, virility

INDIA
purity, love, sensuality, fertility, beauty, wealth, power, wedding color, married woman, fear, fire

ITALY
light, fidelity

JAPAN
life, fertility, passion, strength, anger, danger, self-sacrifice

S. KOREA
good fortune, passion

NETHERLANDS
nature, government, royalty

RUSSIA
communism, revolution, wedding color

THAILAND
Buddhism, color for Sunday

TURKEY
death

ORANGE

WESTERN
energy, vitality, excitement, adventure, creativity, caution, construction, harvest, autumn, affordability, Halloween (with black)

EASTERN
love, happiness, spirituality, humility, good health, immortality

EUROPE
purity, cleanliness, good

MIDDLE EAST
mourning, loss

AUSTRALIA
animals

BRAZIL
environment

FRANCE
earth

INDIA
the most sacred color, purity, courage, sacrifice, religious abstinence, death, rebellion

IRELAND
religious color for Protestants, appears on the Irish flag along with white for peace & green for Catholics

JAPAN
love, happiness, courage

NETHERLANDS
royalty, color of the Dutch royal family

THAILAND
color for Thursday

BROWN

WESTERN
comfort, stability, practicality, wholesomeness, dullness, fertility, organic, earthiness, poverty

EASTERN
earth, mourning

EUROPE
masculinity, earth

MIDDLE EAST
earth, comfort

AFRICA
earth

AUSTRALIA
color of the land

BRAZIL
nature

CHINA
earth

INDIA
color of mourning

INDONESIA
earth

JAPAN
earth, strength, durability

NICARAGUA
disapproval

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Source: Six Degrees. https://www.six-degrees.com/pdf/International-Color-Symbolism-Chart.pdf

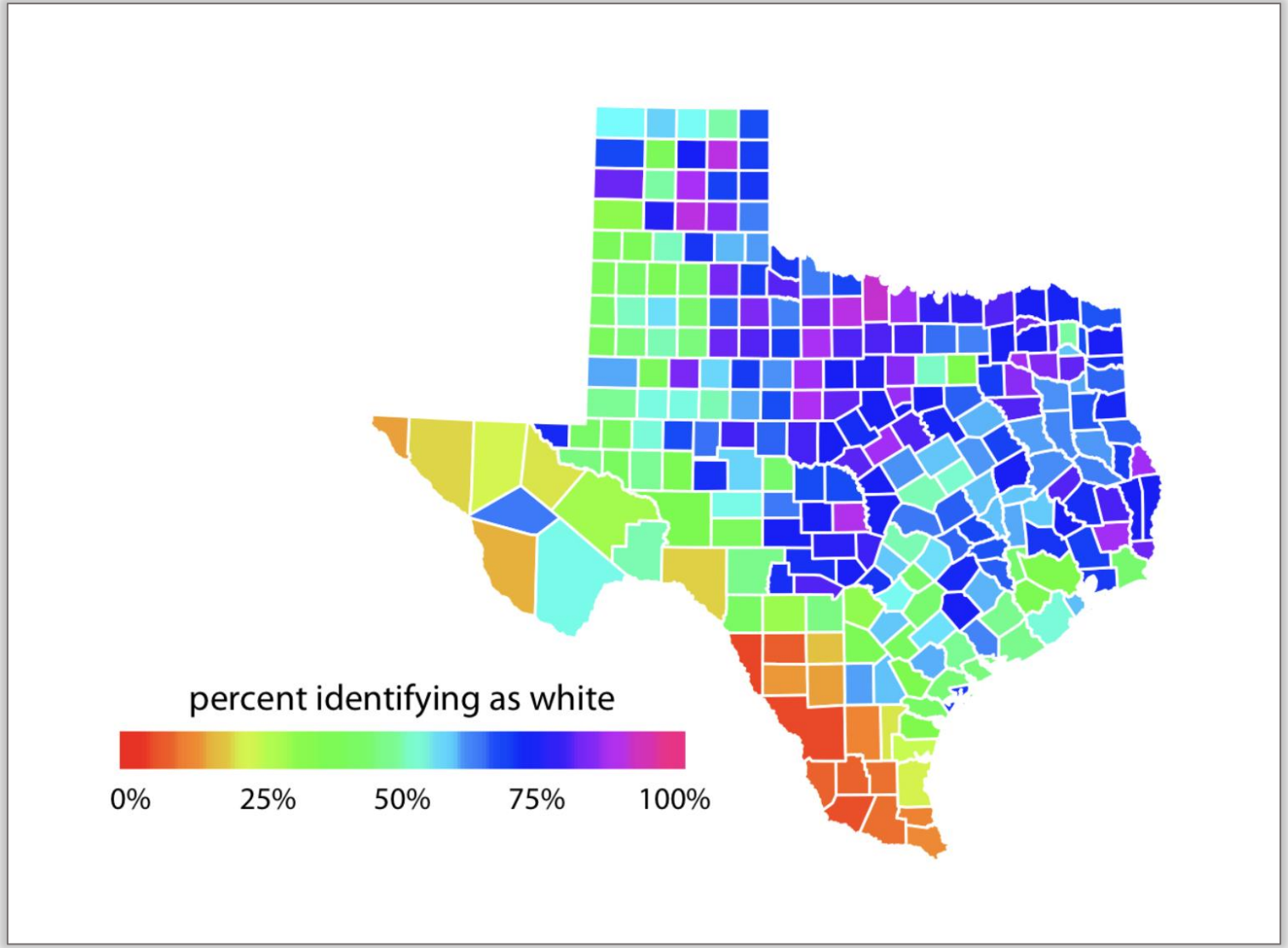
DO – Consider Color Psychology



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DON'T – Use Too Many Colors



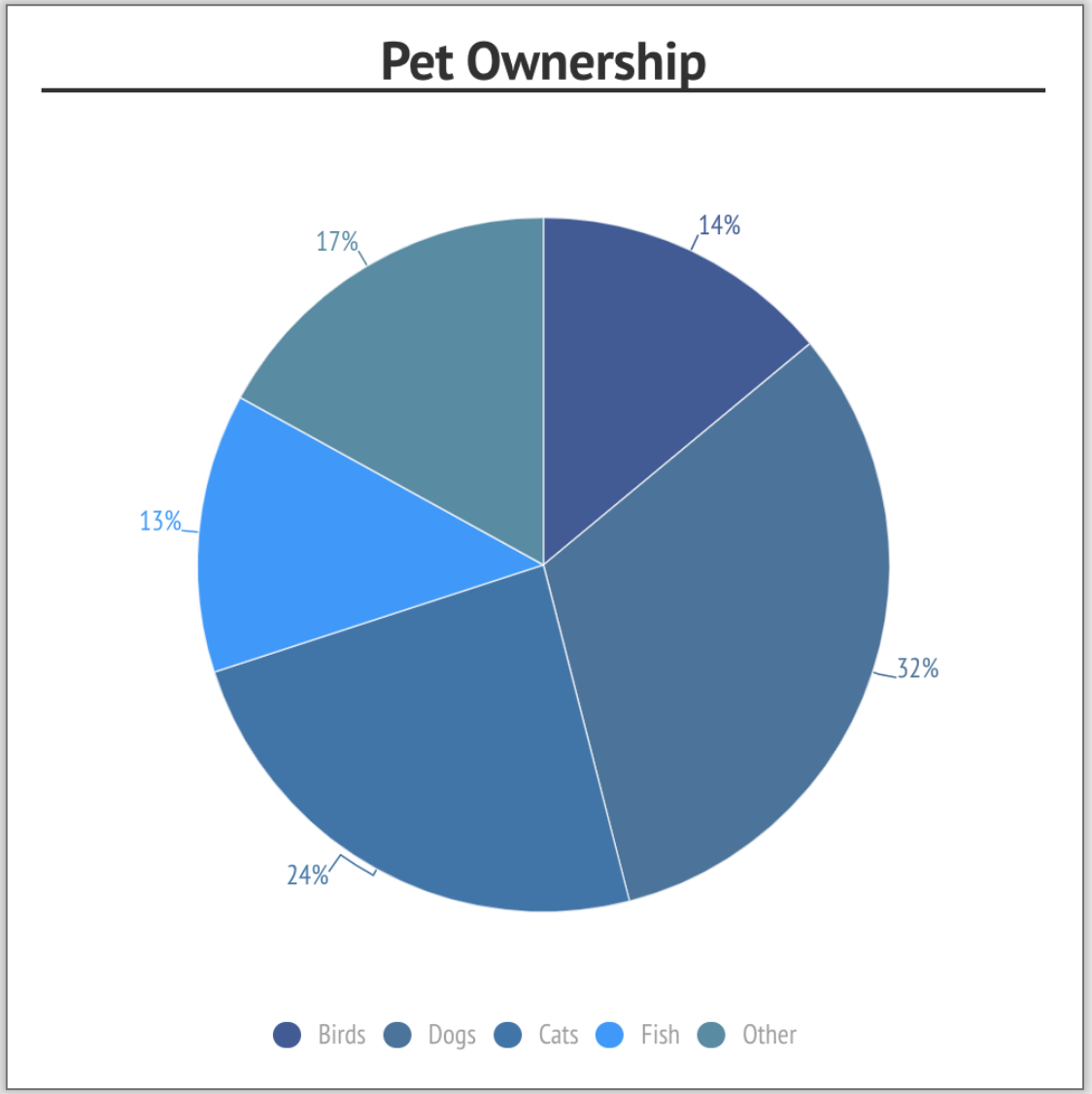
infogram

towards data science

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DON'T – Use Colors too Hard to Distinguish



infogram

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DON'T – Forget Color Blindness

Avoid…

- green + red
- green + brown
- blue + purple
- green + blue
- light green + yellow
- blue + grey
- green + grey
- green + black



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DON'T – Forget Color Blindness

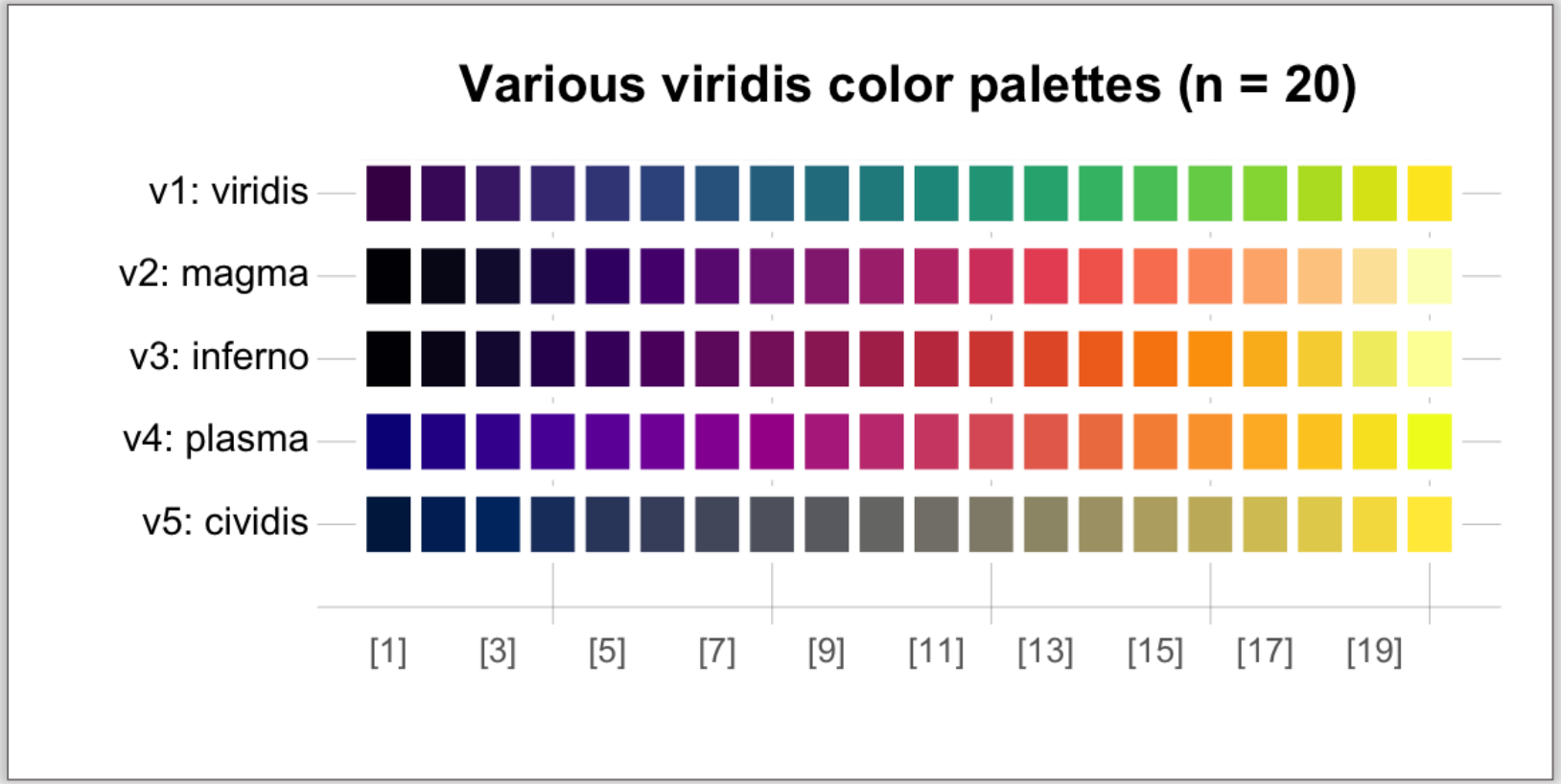
Solution?

VIRIDIS

color library

Make plots that are…

- Elegant
- Better represent your data
- Easier to read for colorblindness
- Print well in greyscale



https://cran.r-project.org/web/packages/viridis/vignettes/intro-to-viridis.html
https://www.geeksforgeeks.org/matplotlib-pyplot-viridis-in-python/

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DON'T – Forget Color Blindness

Instead of green ↑ and red ↓

Use blue ↑ and yellow ↓



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Color Theory Summary



DO

- Keep in mind appropriate uses of color for sequential, diverging, categorical, highlight and alert purposes
- Use light to dark for continuous data
- Use contrasting colors for comparison
- Use your brand colors
- Include a color key
- Use semantic color association
- Consider color psychology



DON'T

- Use too many colors
- Use colors too hard to distinguish
- Forget color blindness

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(homework review)

Data → to → Viz

The Right Graph for your Data

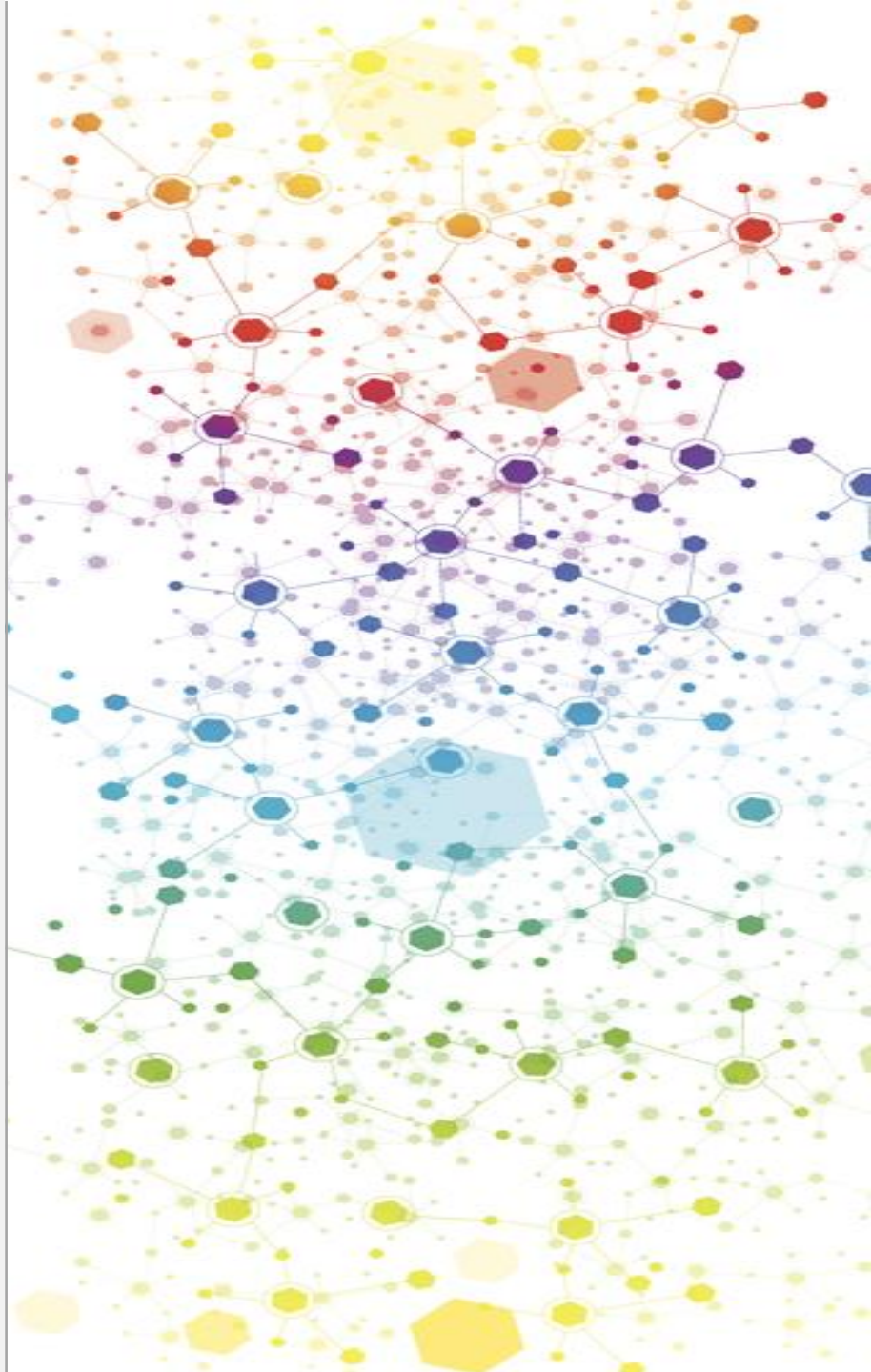


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Taxonomy Intro

How to choose the best graph for:

- Comparing categories
- Showing parts of a whole
- Analyzing temporal changes
- Making connections & relationships
- Mapping geospatial data



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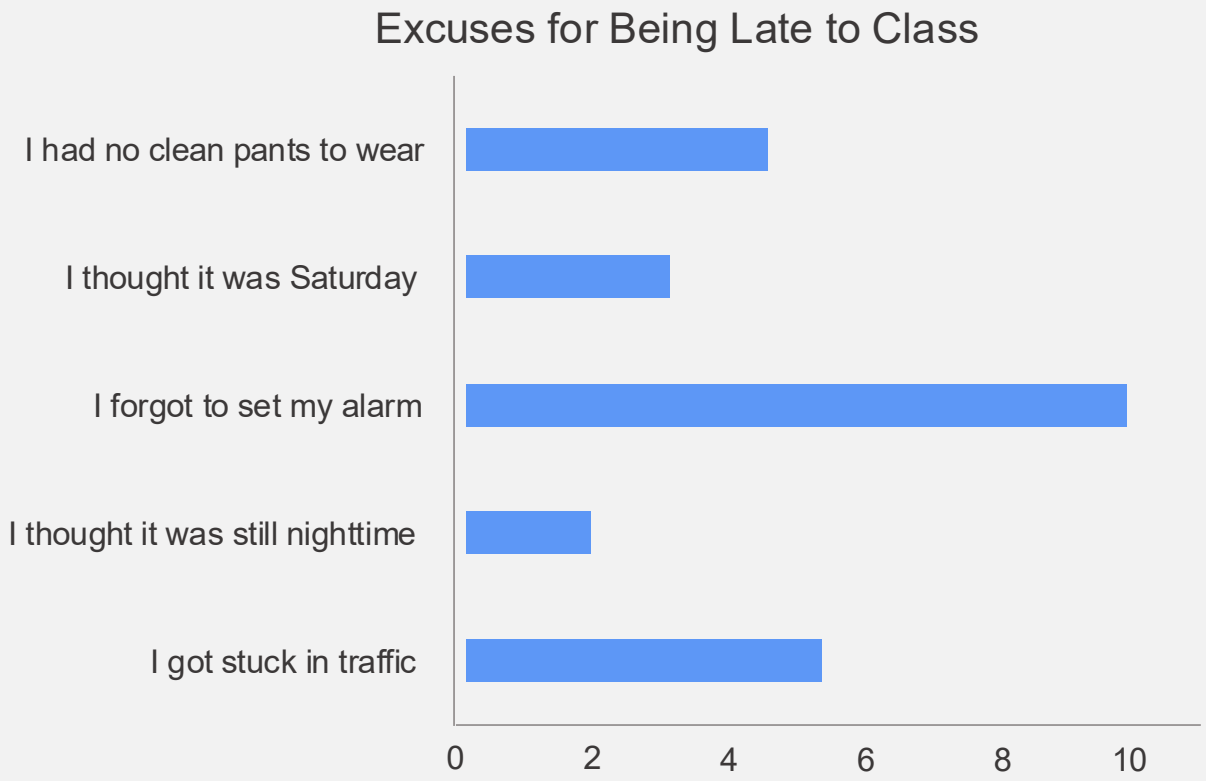
Comparison – Bar Chart

Variables:

- 1 categorical
- 1 quantitative

Great for…

✓Comparing across categories
✓Highlighting differences
✓Showing trends & outliers
✓Revealing highs and lows



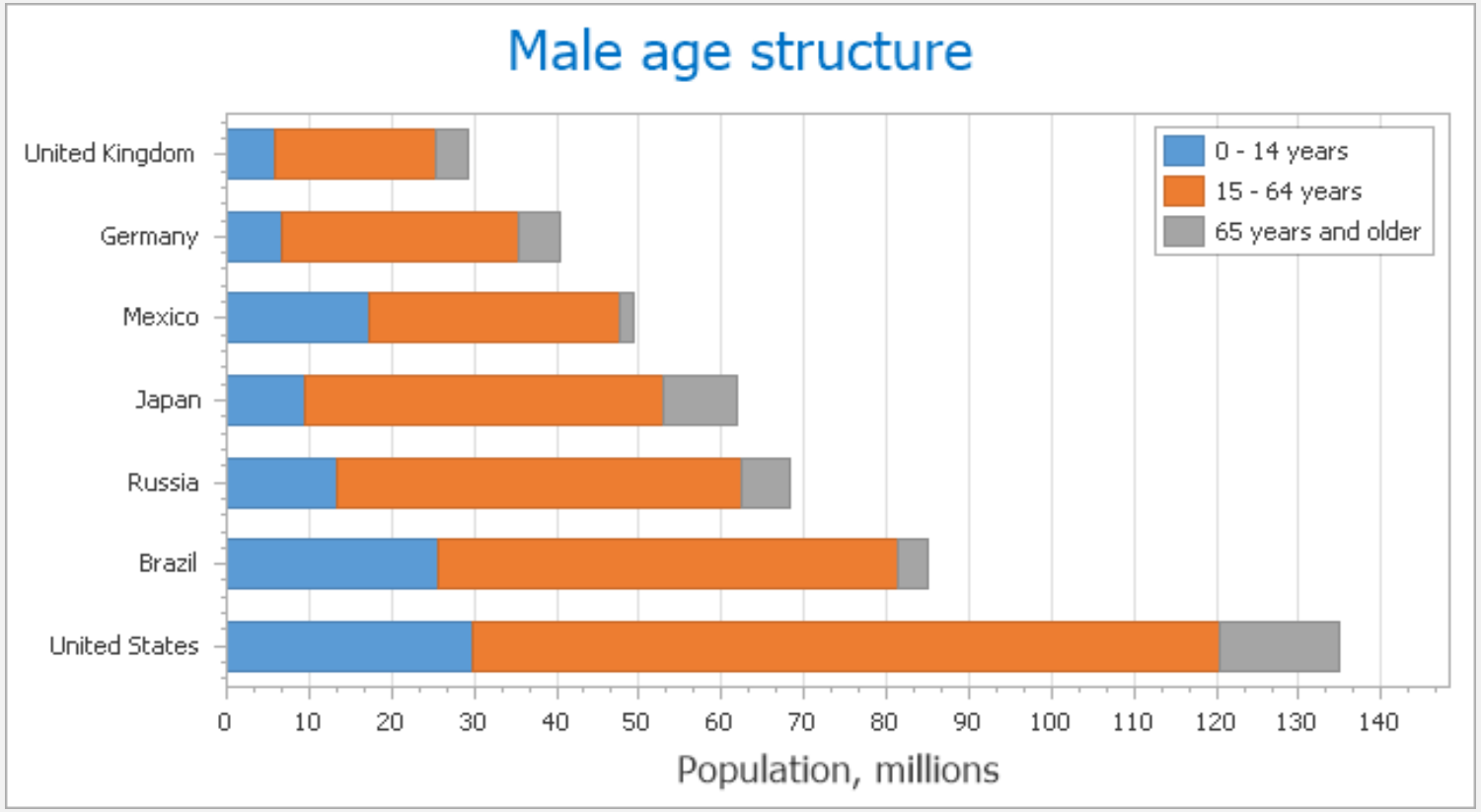
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Comparison – Stacked Bar Chart

Variables:

- 2 categorical (color, position)
- 1 quantitative



Great for…

✓ Highlighting differences

✓ Showing how the total differs over time, and how each piece contributes to the total

Careful…

✓ Bad for showing growth of each variable within (other than the baseline / first variable)

(value annotations can help)

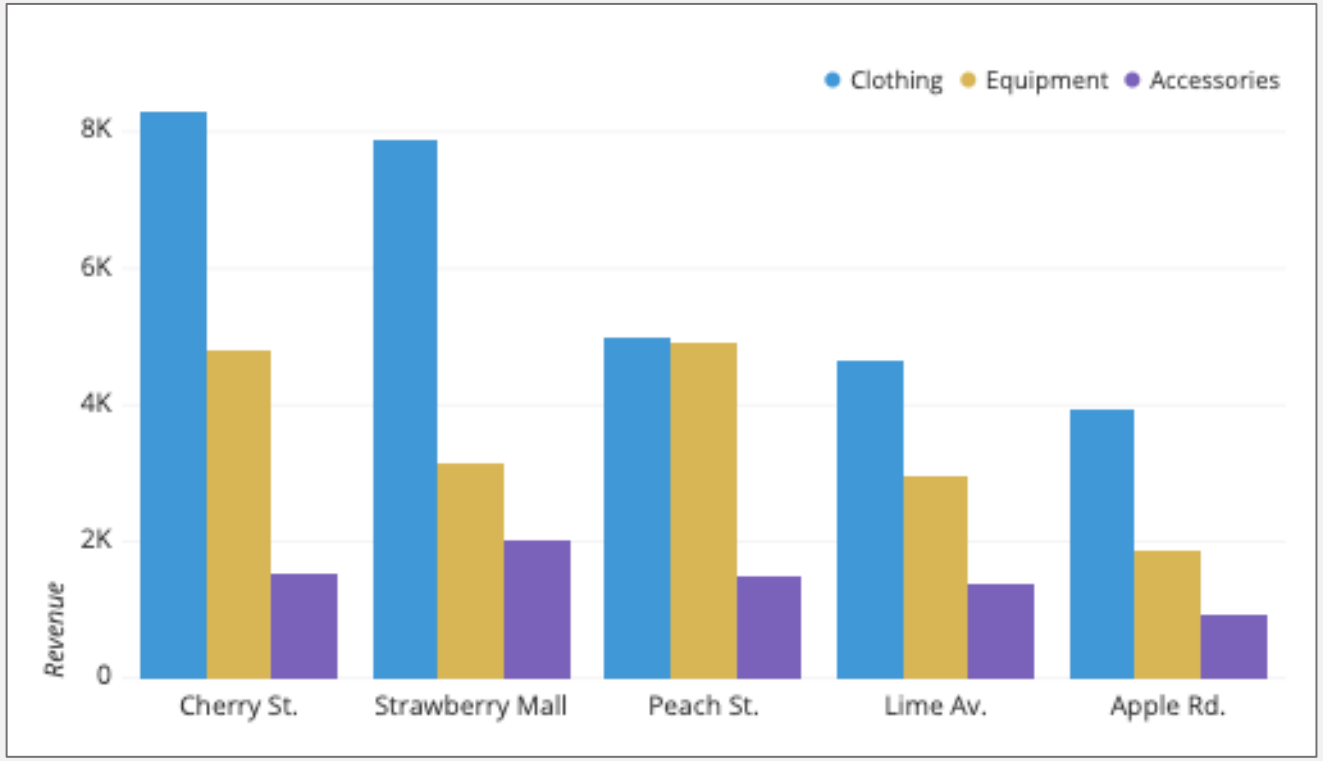
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Comparison – Grouped/Clustered Bar Chart

Variables:

- 2 categorical (color, position)
- 1 quantitative



Great for…

✓ Highlighting differences

✓ Precise understanding of how each section / category compares

Careful…

✓ No way to observe what each category represents of total

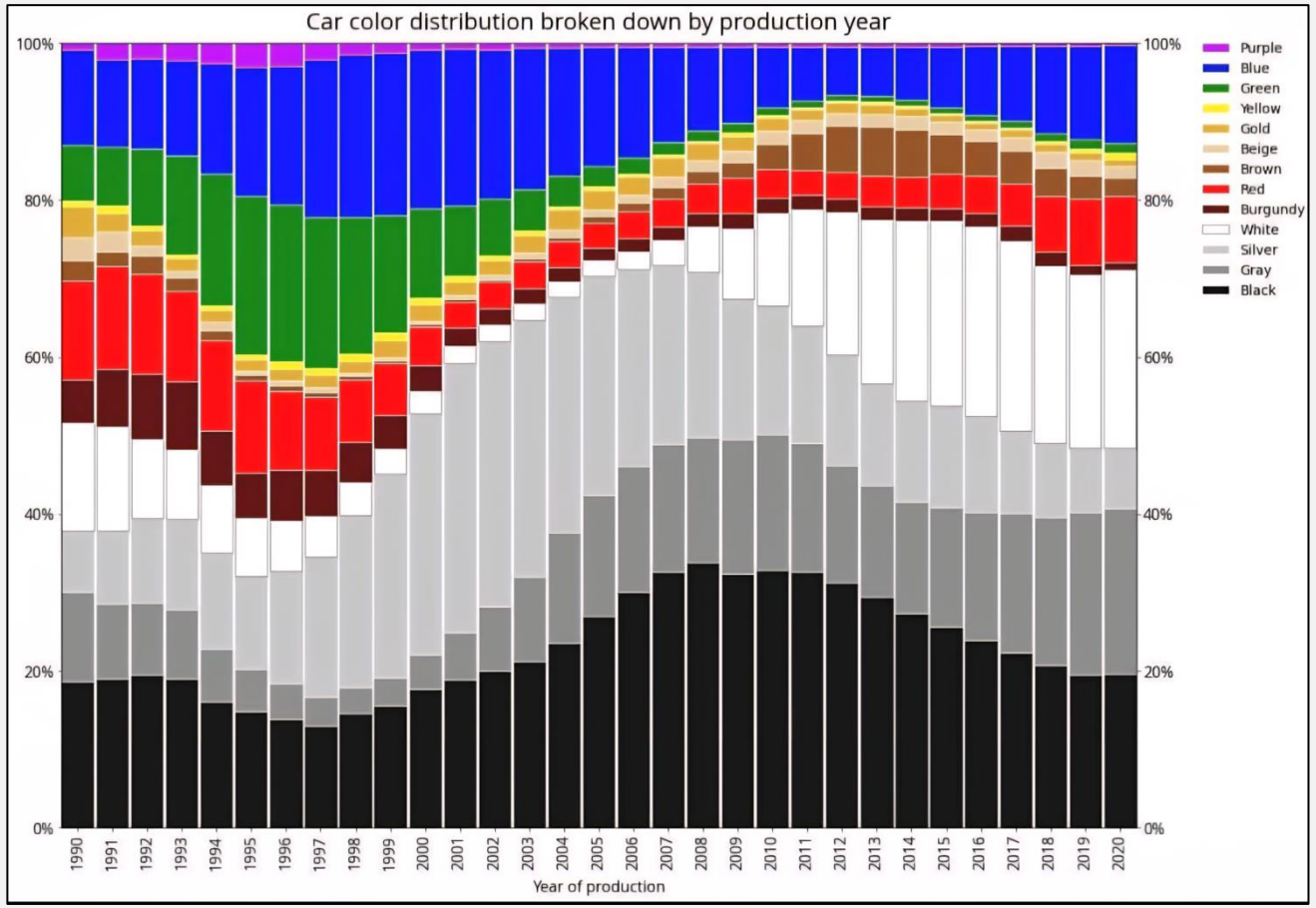
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Comparison – Stacked Cumulative Bar Chart

Variables:

- 2 categorical (color, position)
- 1 quantitative



Great for…

- ✓ Highlighting how the distribution changes over time
- ✓ Showing how each piece contributes to the total

Careful…

- ✓ Bad for showing growth of each variable within (other than the two baselines at each end)
- ✓ No way to understand how totals differ over time

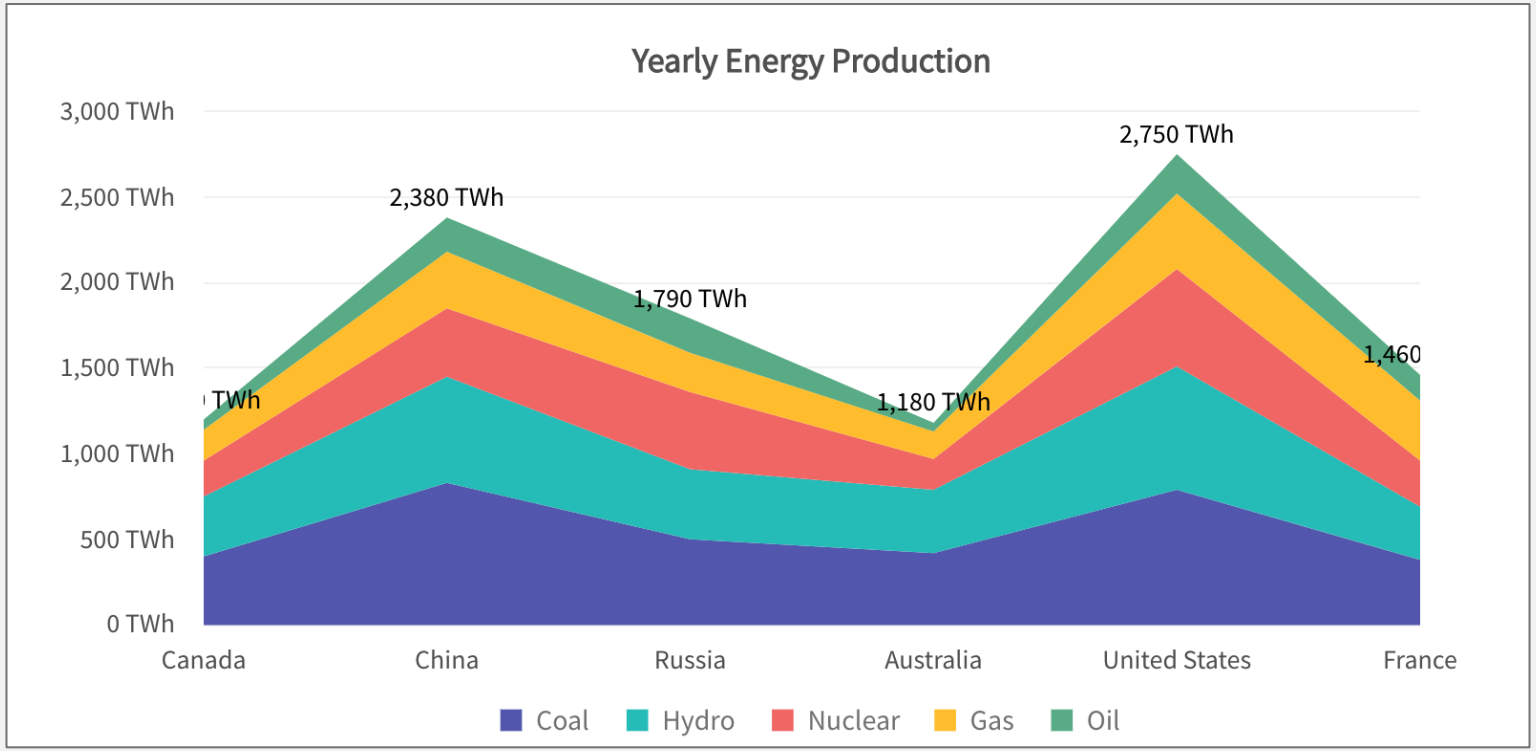
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Comparison – Area Chart *(same concept as Stacked Bar Chart!)*

Variables:

- 2 categorical
- 1 quantitative



Great for…

✓ Highlighting differences

✓ Showing how the total differs over time, and how each piece contributes to the total

Careful…

✓ Bad for showing growth of each variable within (other than the baseline / first variable)

(value annotations can help)

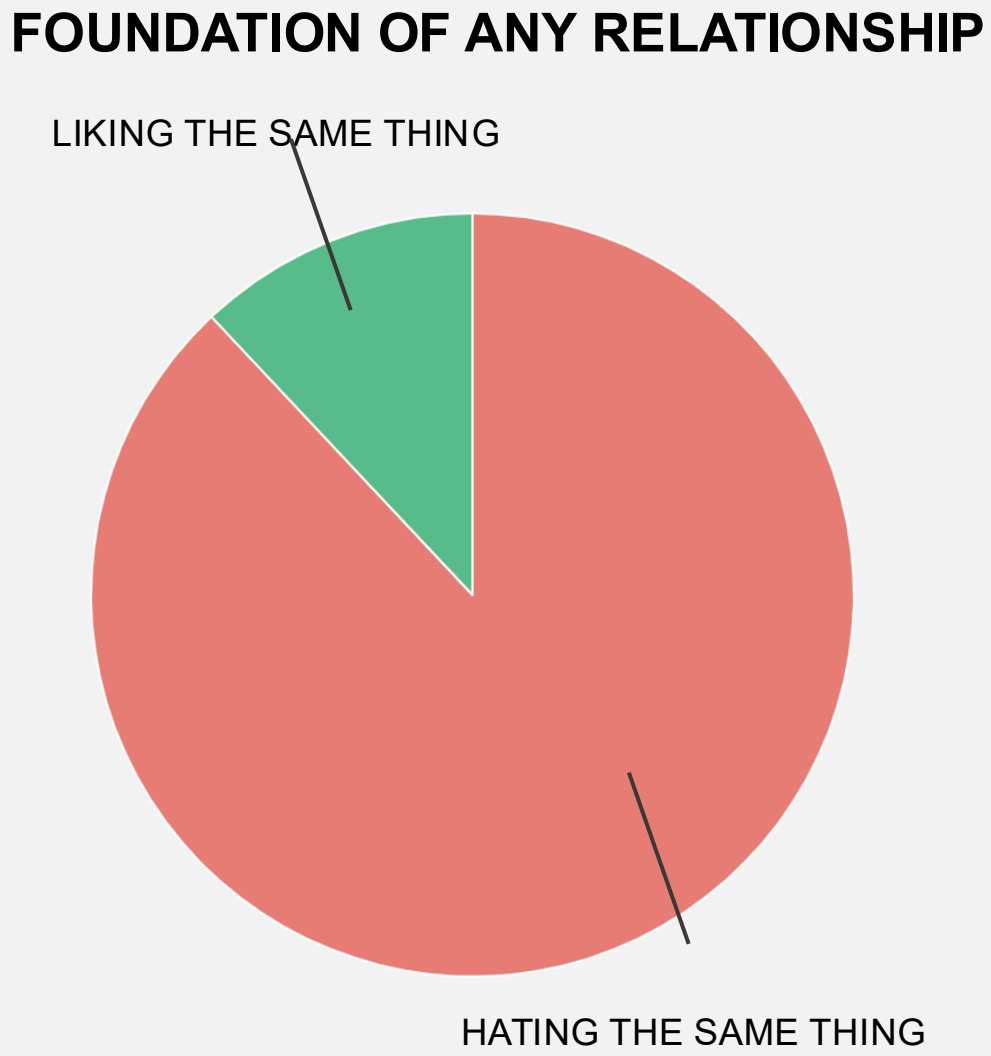
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Parts of a Whole – Pie Chart

Variables:

- 1 categorical (color)
- 1 quantitative-ratio (angle)



Great for…

✓ Comparing across few categories, each with significantly varying values

Careful…

✓ Bad for many categories

✓ Bad for several categories with similar values

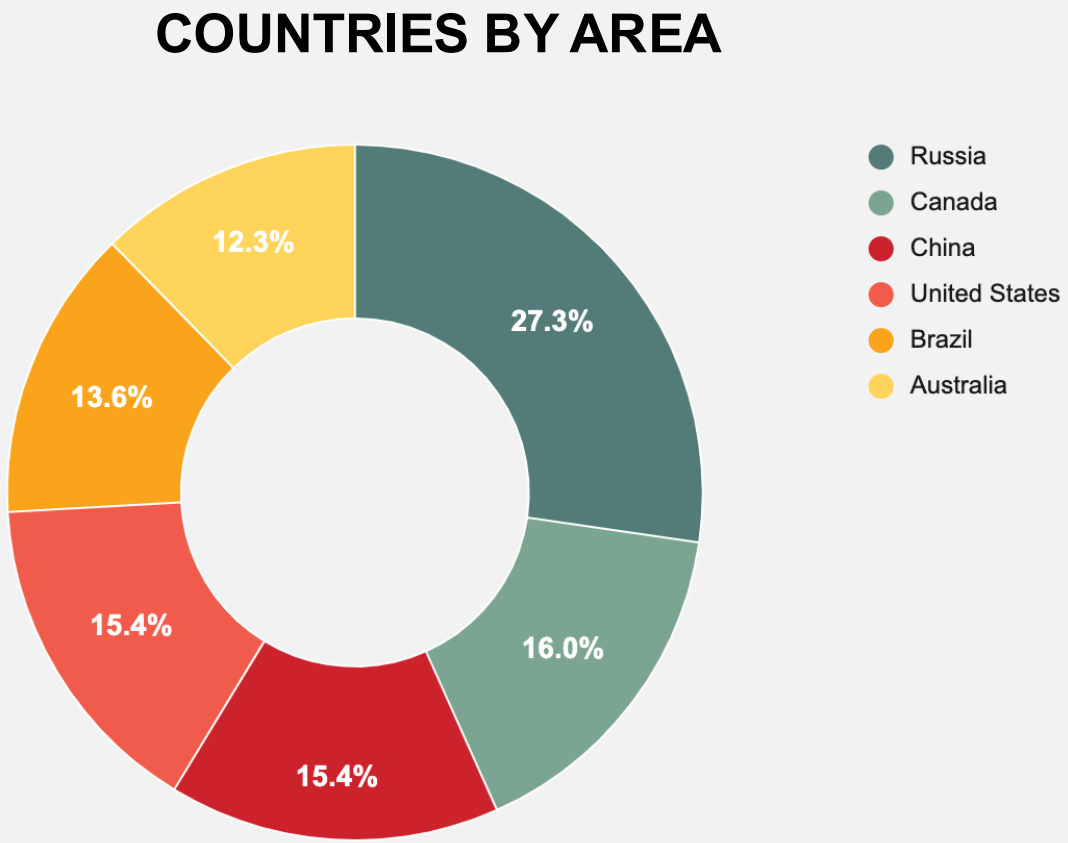
(value annotations can help)

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Parts of a Whole – Donut Chart *(skip the Pie Chart!)*

Variables:
- 1 categorical (color)
- 1 quantitative-ratio (angle)



Great for…
✓Comparing information across categories in a simplistic way

Careful…
✓Bad for many categories
✓Bad for several categories with similar values
(value annotations can help)

★**TIP:** Always use a separator (in my case here, white borders)

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Parts of a Whole – Pie Chart | Donut Chart



Why so much hate?

→ Less accurate
→ Harder to interpret.

People are much better at comparing length or height, rather than areas.

Thinner pieces become unreadable.

Difficult to understand the difference between pie slices with similar values.

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Parts of a Whole – Circle Packing Diagram

Variables:

- Multiple categorical (color, position)
- 1 quantitative-ratio (area)



Great for…

✓ Showing distribution and part-to-whole in very large datasets

Careful…

✓ Don't over-do, it's typically used more for 'show' or the wow factor

✓ Uses space less efficiently than a treemap

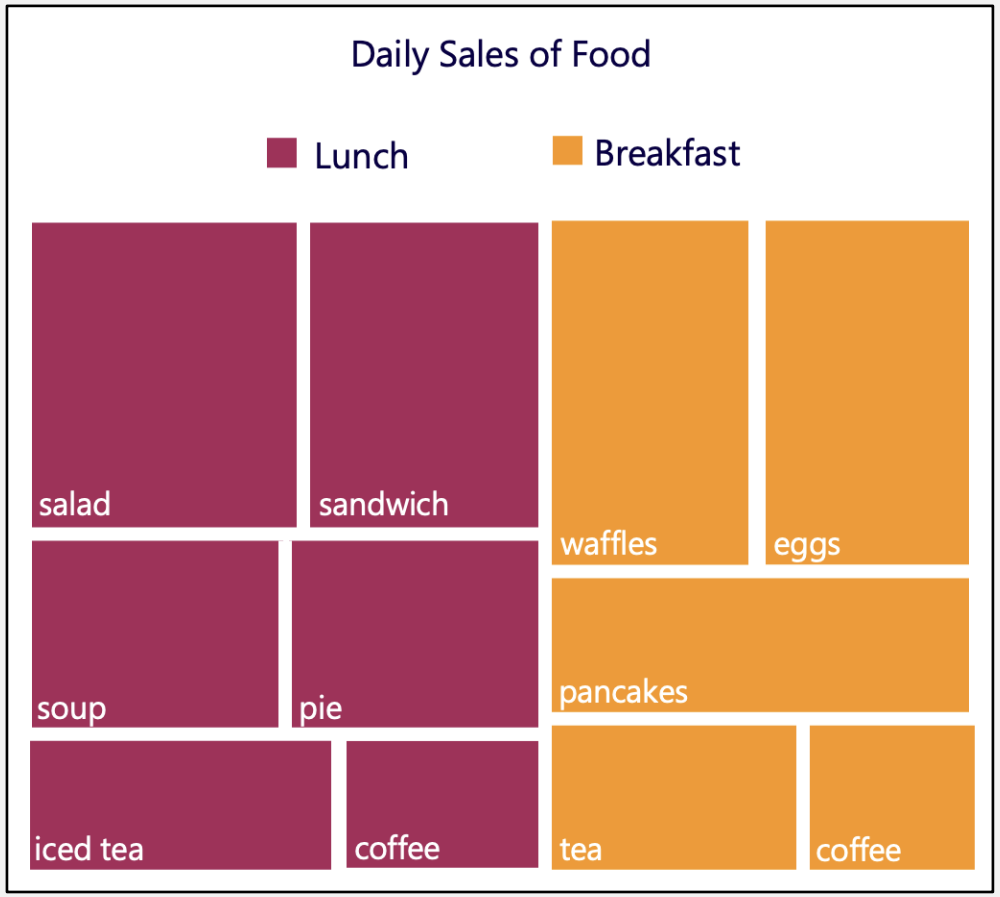
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Parts of a Whole – Treemap

Variables:

- Multiple categorical (color, position)
- 1 quantitative-ratio (area)



Great for…

✓ Showing distribution and part-to-whole in very large datasets

✓ Uses space more efficiently than a circle packing diagram

Careful…

✓ Don't over-do it - too many categories, especially small ones, can quickly get overwhelming

✓ Category names can easily get cut off and thus become unreadable

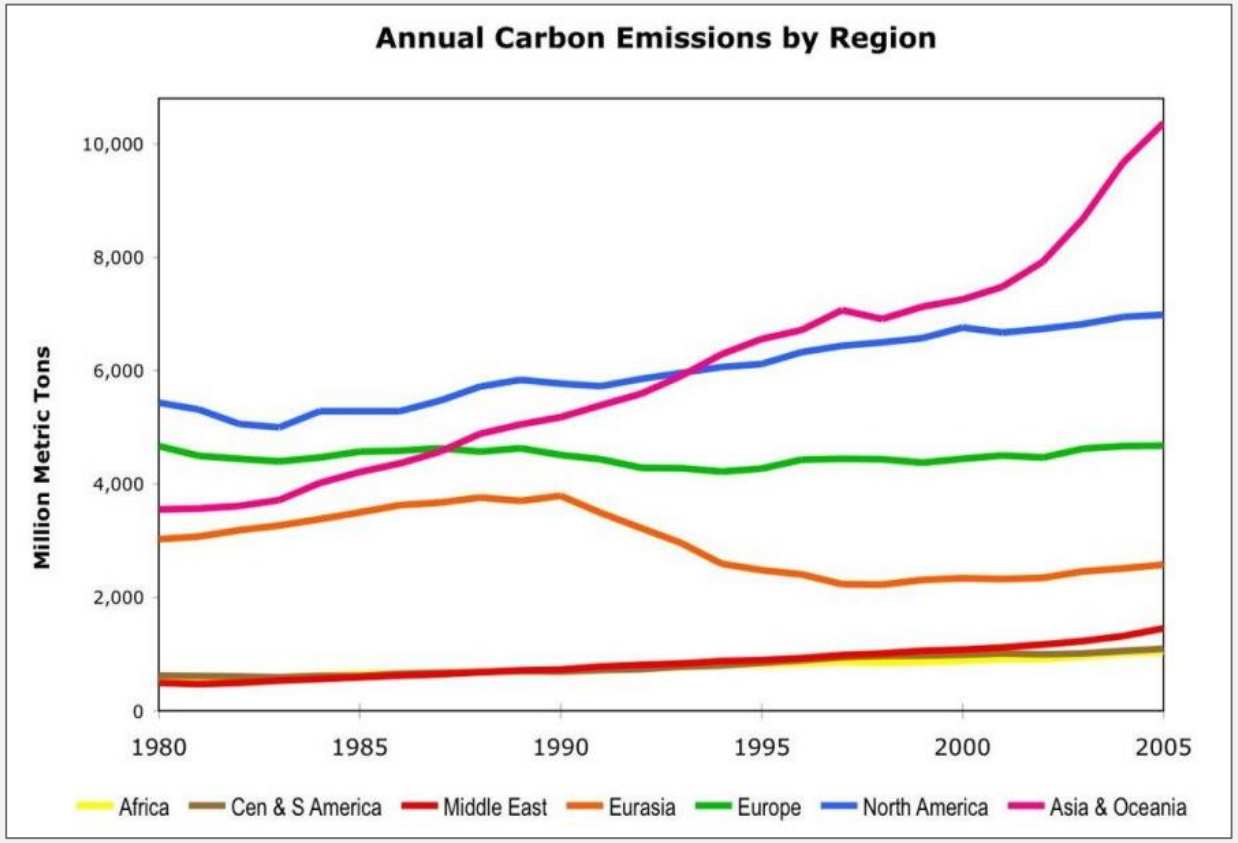
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Time Series – Line Chart

Variables:

- 1 categorical (color)
- 1 quantitative-interval
- 1 quantitative-ratio



Great for…

✓ Viewing trends, usually over time (can also be over time intervals and other ordinal data)

Careful…

✓ Use too many lines and it becomes a 'spaghetti chart'

★**TIP:** Combine a line graph with a bar chart, they go well together if used correctly!

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Time Series – Line Chart

Variables:

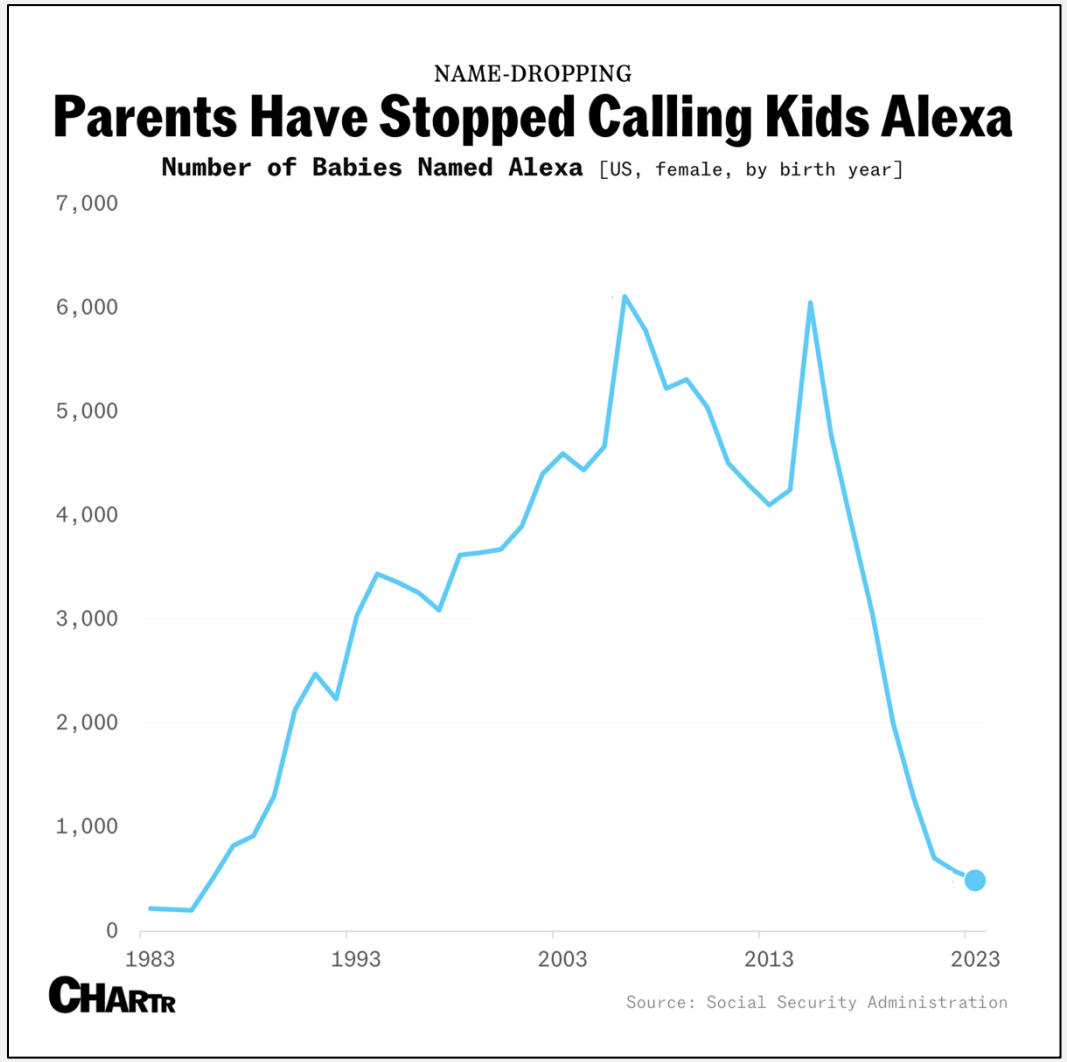
- 1 categorical (color)
- 1 quantitative-interval
- 1 quantitative-ratio

Great for…

✓ Viewing trends, usually over time (can also be over time intervals and other ordinal data)

Careful…

✓ Use too many lines and it becomes a ‘spaghetti chart’



★**TIP 2:** Use annotations to tell a story!

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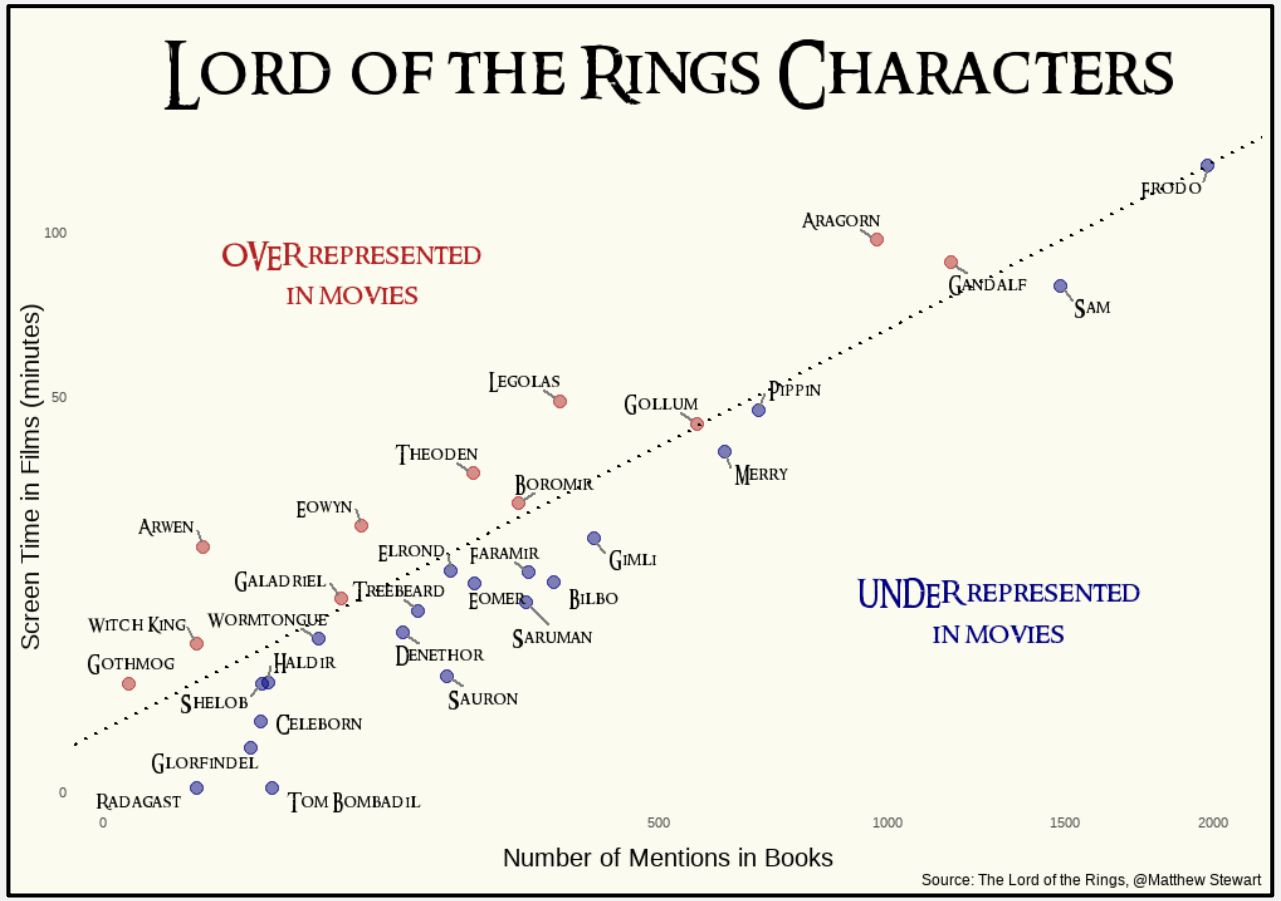
Relationships – Scatterplot

Variables:

- 2 quantitative

Great for…

✓ Investigating a relationship between different variables
✓ Showing correlations
✓ Often used in regression analysis



★**TIP:** Add a trend line to show correlation

★**TIP:** Customize points to highlight where you want the user to focus

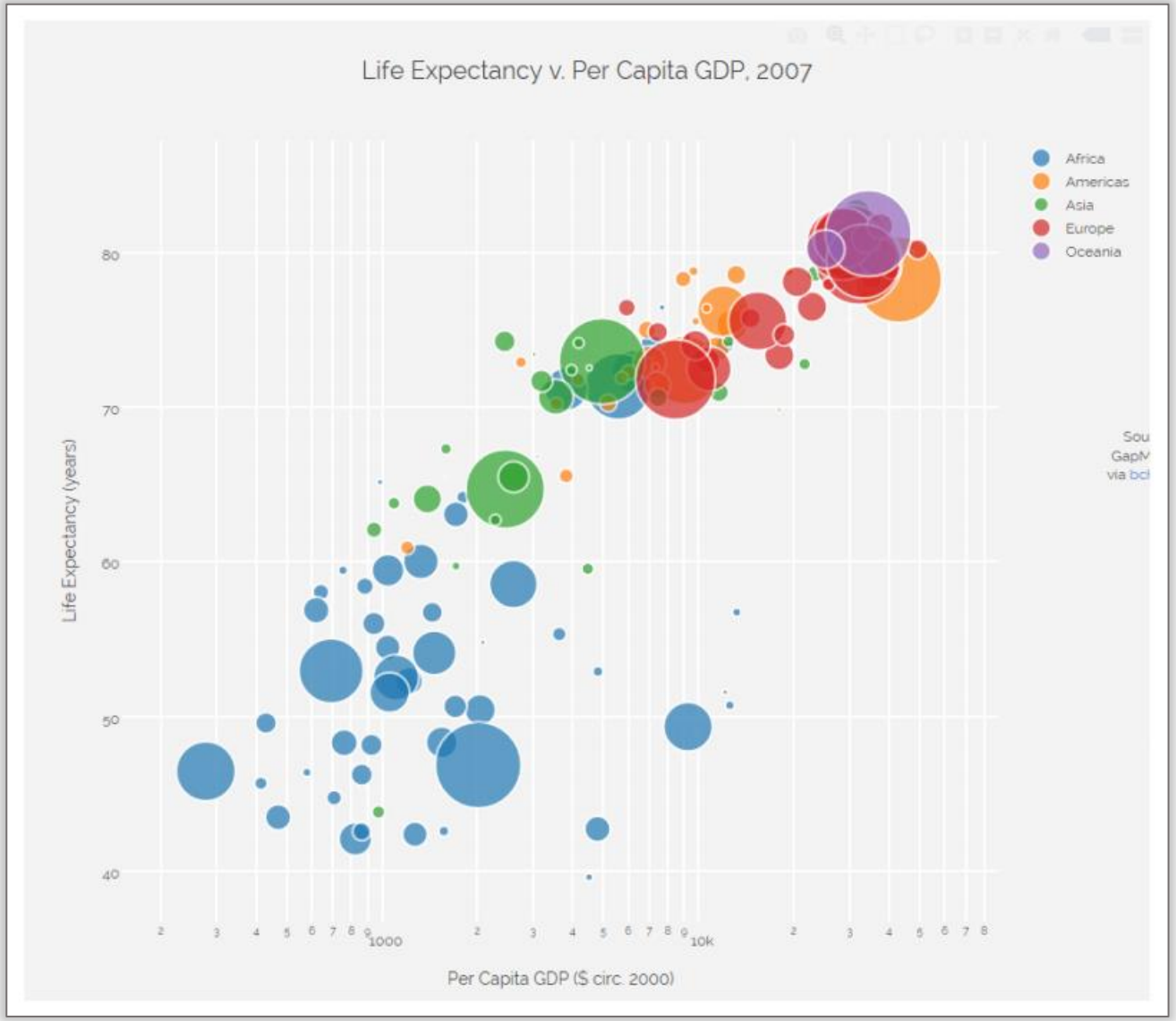
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Relationships – Bubble Plot

Variables:

- 2-3 quantitative
- 1-2 categorical



Great for…

✓ Showing large amounts of information at once

★**TIP:** Use color as an extra dimension to visualize an additional category

★**TIP:** Overlay on a map when using geographically-related data

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Relationships – Heat Map

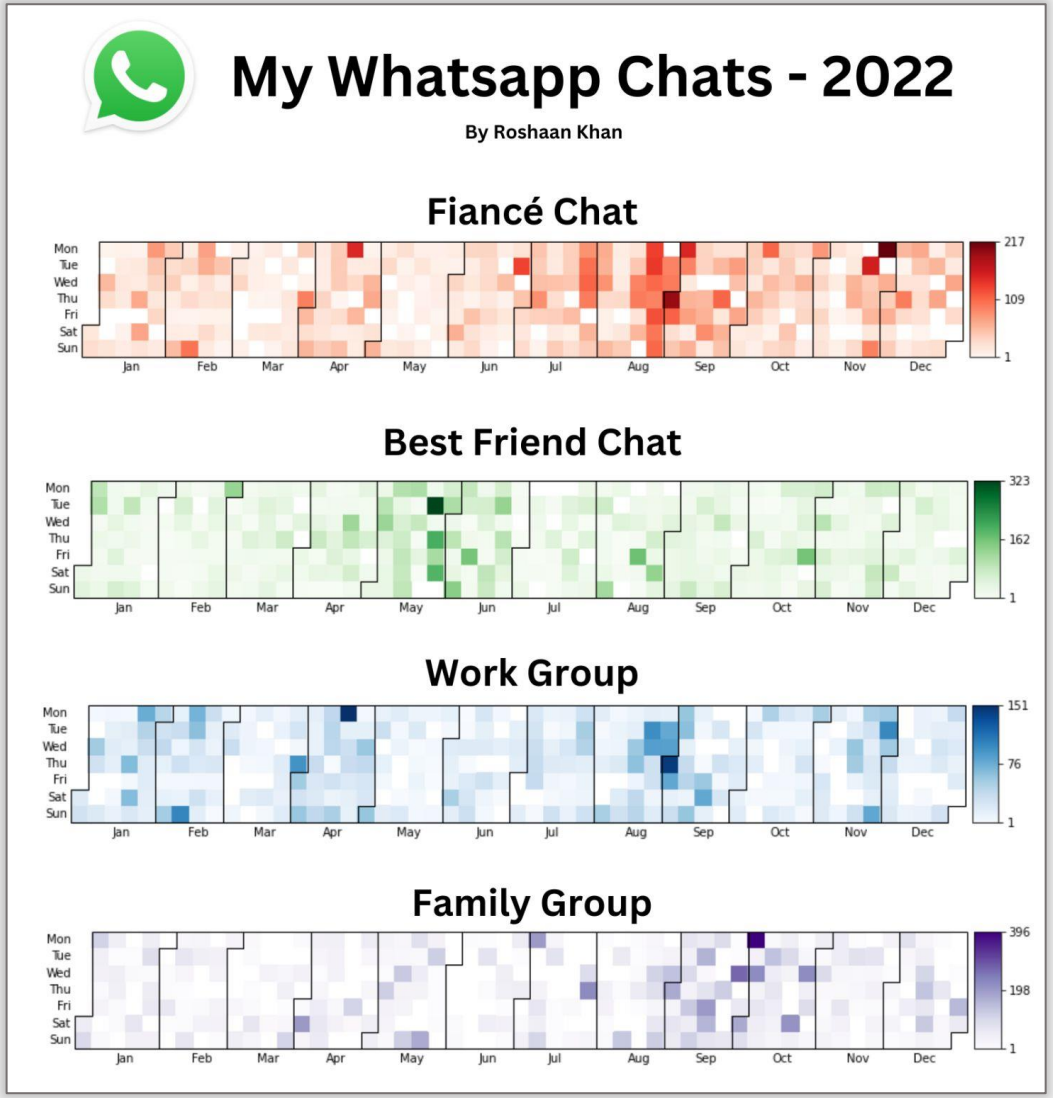
Variables:

- 1 quantitative
- Multiple categorical (x and y axis)

Great for…

✓ Using color to highlight the progress of data points, usually over time

★**TIP:** Use a 'highlight table' that includes the precise figures when needed



Source: Roshaan Khan

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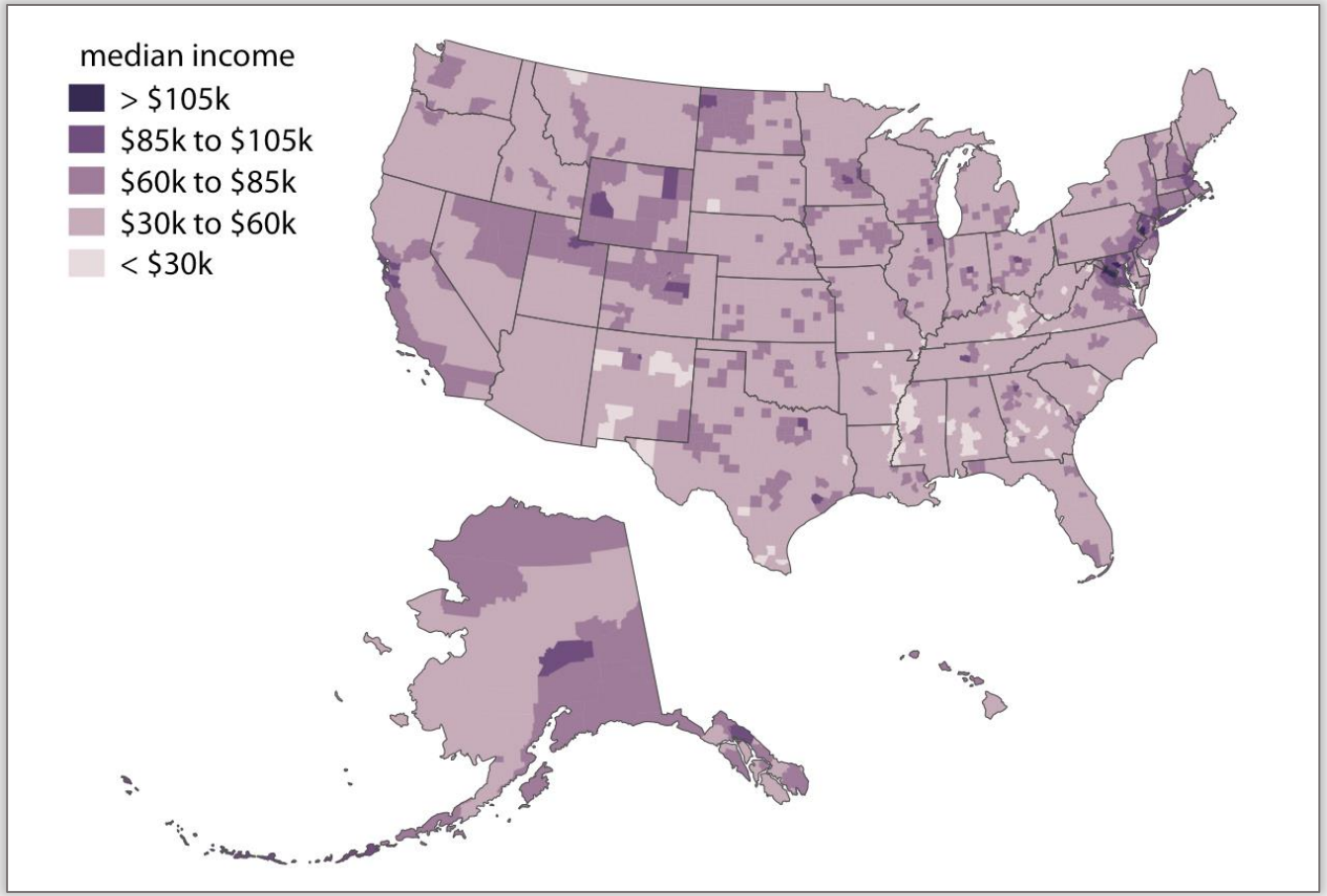
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Source: Claus O Wilke. https://clauswilke.com/dataviz/geospatial-data.html, Original Data Source: 2015 Five-Year American Community Survey

Mapping – (Geospatial) Discrete Choropleth Map

Variables:

- Geo data



Great for…

✓Visualizing data connected to country names, postal codes, states, etc

Careful…

✓You may need to take population / density into account

★**TIP:** Continuous scales look visually appealing but harder to read, stick to bucketed discrete groups (although loses some detail)

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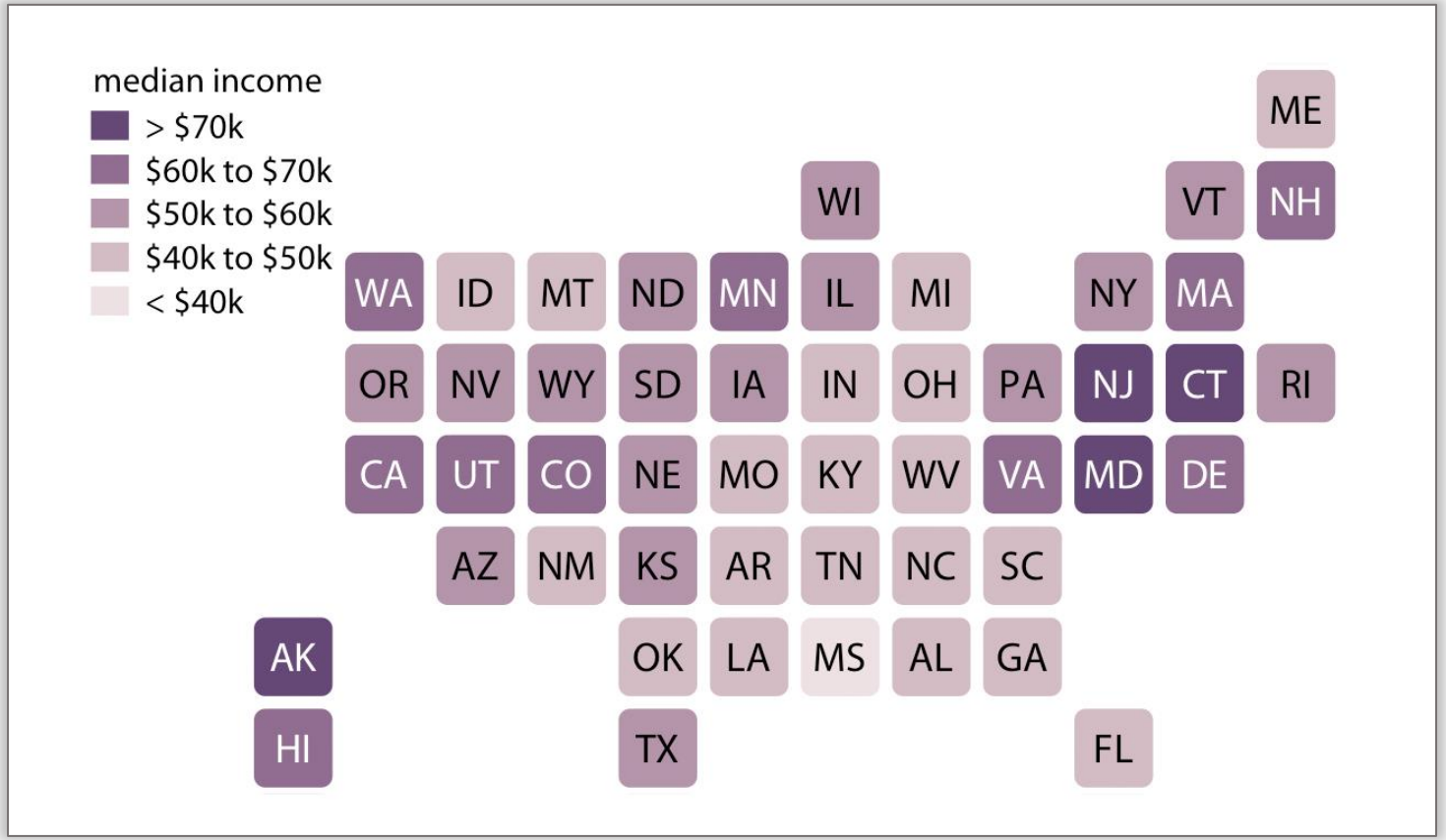
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Source: Claus O Wilke. https://clauswilke.com/dataviz/geospatial-data.html, Original Data Source: 2015 Five-Year American Community Survey

Mapping – (Geospatial) Cartogram Heatmap

Variables:

- Geo data



Great for…

✓ Visualizing data connected to country names, postal codes, states, etc

Careful…

✓ Does not correct for population or consider size, treats all geos equally

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It’s a lot to remember, isn’t it?



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from Data to Viz

From Data to Viz leads you to the most appropriate graph for your data. It links to the code to build it and lists common caveats you should avoid.



https://www.data-to-viz.com/

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matplotlib seaborn plotly

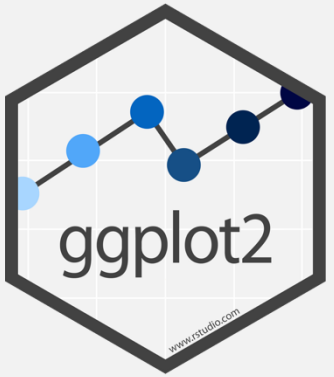
The Python Graph Gallery

Welcome to the Python Graph Gallery, a collection of hundreds of charts made with `Python`. Charts are organized in about 40 sections and always come with their associated reproducible code. They are mostly made with `Matplotlib` and `Seaborn` but other library like `Plotly` are sometimes used.

https://www.python-graph-gallery.com/

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The R Graph Gallery

Welcome the R graph gallery, a collection of charts made with the R programming language. Hundreds of charts are displayed in several sections, always with their reproducible code available. The gallery makes a focus on the tidyverse and ggplot2

https://r-graph-gallery.com

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Feel free to stay in touch!

(I've reached the 30K connection limit on LinkedIn unfortunately, so until I can get that back down, you can follow me there!)

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